Who Annotates in NLP? A Large-scale Assessment of Human Annotation Reporting between 2018 and 2025

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Abstract

Human annotation is the empirical foundation of much NLP research, from dataset construction to model evaluation, but papers often leave it unclear who produced the annotations and how the annotation process was controlled. We provide the first large-scale, task-level audit of human annotation reporting across major NLP venues, asking which annotation details are documented, which are missing, and how reporting varies across time, topic, venue, and intended use of human judgment. We introduce a unified taxonomy of annotation-reporting practices and validate an LLM-assisted extraction pipeline against ANNOTATED$_{\text{GOLD}}$, a human-adjudicated gold standard of 41 papers and 72 annotation tasks. Our best LLM model reaches human-comparable agreement with adjudicated labels (Krippendorff's $\alpha = 0.606$ vs. 0.585 for human–human agreement). Using this pipeline, we construct ANNOTATED$_{\text{LLM}}$, a dataset covering ACL-venue papers from 2018–2025, with 2,667 extracted annotation tasks from 1,603 papers. We find that papers frequently report operational details such as recruitment strategies, annotator expertise, and annotation volume, but often omit details needed to assess annotation validity, including training, language proficiency, compensation, socio-demographics, adjudication, and agreement values, especially in model-evaluation studies. Our results show that annotation reporting in NLP has improved over time but remains uneven. Based on these findings, we propose a scalable framework and bare-minimum reporting recommendations for making human annotation more reliable, reproducible, and interpretable.[1]

1 Introduction

Human annotation is one of the core foundations of empirical NLP. When evaluating a machine translation system, building an argument-mining dataset or comparing the outputs of large language models (LLMs), we often rely on human judgments as ground truth. As a result, many NLP claims depend not only on models and metrics, but also on who the annotators are, what they are asked to do, how they are trained, how disagreements are handled (Artstein and Poesio, 2008; van der Lee et al., 2019; Popović and Belz, 2021). Unqualified or biased human annotation may undermine the validity of findings in NLP research: for example, a study evaluating AI-generated poetry may critically depend on annotators' language proficiency, and a study of political stance, sexism, or hate speech may yield different outcomes based on annotators' social and ideological positions.

Worryingly, there is extremely little empirical evidence about the annotators in the NLP community: does the community use crowdworkers or experts (and in what proportions), are the annotators adequate for the task, and how well are they compensated? Further, there is a lack of quantitative evidence on whether initiatives such as the ACL Responsible NLP Checklist (Rolling Review, 2022), introduced to improve the consistency, transparency, and ethics of AI research, correspond to measurable improvements in annotation reporting quality, how these reporting practices differ across intended uses of human annotation, and which NLP areas systematically underreport information critical for interpreting annotation outcomes. We address this gap with the first large-scale, task-level audit of human annotation reporting across major NLP venues.[2]

Existing work has examined annotation quality, reproducibility, annotator effects, and selected aspects of reporting practices (Bayerl and Paul,

[1]Code and data are available at https://github.com/NL2G/who-are-the-annotators.

[2]Our task-level perspective is essential: a single paper may contain multiple annotation tasks with different annotators, instructions, purposes, and quality-control measures. Treating papers as the unit of analysis would therefore obscure task-level variation in annotators, procedures, and quality control that are crucial for evaluating annotation setups.

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2011; Belz et al., 2023; Klie et al., 2024; Beck, 2023). However, prior studies typically focus on specific annotation problems or isolated reporting dimensions, such as inter-annotator agreement (Artstein and Poesio, 2008; Bayerl and Paul, 2011), reproducibility of human evaluation (Belz et al., 2023; Belz and Thomson, 2024), dataset quality management (Klie et al., 2024), or annotator demographics and data documentation (Bender and Friedman, 2018; Pei and Jurgens, 2023). In contrast, we provide the first large-scale, task-level, cross-topic analysis of annotation reporting across major NLP venues, using a unified taxonomy that covers not only agreement and workload, but also recruitment, qualifications, compensation, sociodemographics, and quality control. We also present the first large-scale empirical evidence on how reporting has changed over nearly a decade, starting from 2018, which allows us to examine how reporting practices in the NLP community have changed over time.

Overall, we make the following contributions:

(1) A task-level reporting dataset. We introduce ANNOTATED, a dataset for auditing human annotation reporting in NLP. It treats each annotation task as a separate unit, capturing within-paper differences in annotators, instructions, purposes, and quality-control procedures.

(2) A human-adjudicated gold standard. We manually annotate $\text{ANNOTATED}_{\text{GOLD}}$, a gold-standard set of 41 papers comprising 72 human-annotation tasks, and finalize labels through adjudication. This compact set reflects the substantial cost of task-level expert annotation, estimated at approximately €6,300 in personnel time (Appendix C), while providing a human-validated benchmark for evaluating automated methods that extract annotation-reporting information.

(3) An LLM-assisted extraction pipeline. We evaluate six LLMs as structured extractors of annotation-reporting information. The strongest model, Gemini-3.1-Pro, reaches Krippendorff's $\alpha = 0.606$ against the adjudicated gold standard; a separate comparison against individual pre-adjudication human annotations further supports human-comparable extraction performance.

(4) The first large-scale audit of annotation reporting in ACL-venue papers. We apply the validated pipeline to construct $\text{ANNOTATED}_{\text{LLM}}$, a corpus of ACL-venue papers published between 2018 and 2025 and extract 2,667 annotation tasks from 1,603 papers. We analyze reporting practices over time, venues, NLP topics, and uses of human judgment, including resource creation, model evaluation, and human-performance benchmarking.

(5) A scalable framework for more reliable annotation reporting. Motivated by our large-scale audit of annotation-reporting practices, we propose a scalable framework and bare-minimum reporting recommendations for making human annotation more reliable, reproducible, and interpretable (§7). Since NLP models, benchmarks, and datasets often rely on human annotation for validation, improving annotation reporting has foundational implications for the interpretation of empirical claims in NLP.

2 Related Work

Annotator effects, annotation quality, and reporting. Human annotation has long been recognized as central to NLP evaluation and dataset construction, with prior work emphasizing the importance of agreement measurement, evaluation design, and reproducibility (Artstein and Poesio, 2008; van der Lee et al., 2019; Popović and Belz, 2021). A growing body of research further shows that annotators are not interchangeable: their demographic backgrounds, language competence, cultural context, and social or political attitudes can systematically affect annotation outcomes. For instance, Pei and Jurgens (2023) show that demographic variables influence offensiveness judgments, while Jiang et al. (2024) find that annotators' social and political attitudes affect sexism and misogyny annotations. Related work highlights the need to document demographic and linguistic context in NLP datasets and evaluations (Bender and Friedman, 2018; Joshi et al., 2020). At the same time, work on reproducibility has shown that many human evaluations cannot be fully reproduced because papers omit essential information about annotators, materials, instructions, and procedures (Belz et al., 2023; Belz and Thomson, 2024). Earlier meta-analytic work examined annotation and annotator characteristics across 96 studies (Bayerl and Paul, 2011), annotated-data quality dimensions (Beck, 2023), and annotation quality management in dataset papers (Klie et al., 2024).

AI4Science and meta-scientific analysis. AI4Science studies how AI systems can support or transform scientific work, including literature analysis, discovery, experimentation, writing, figure generation, and evaluation (Pramanick et al., 2025; Chen et al., 2025b; Eger et al., 2026;

Xie et al., 2025). Recent work has explored LLM-based support for peer review (Tyser et al., 2024), abstract generation (Kim et al., 2025), and scientific graphics generation (Belouadi et al., 2025; Greisinger and Eger, 2026). In NLP, LLMs have also increasingly been used as annotators, evaluators, or assistants for scientific and empirical analysis, enabled by the general capabilities of LLMs (Brown et al., 2020; Grattafiori et al., 2024; Singh et al., 2026). For example, LLMs have been used to support annotation in social-scientific analysis (Kostikova et al., 2024; Ziems et al., 2024) to interpret latent semantic spaces learnt in pretrained models (Mousi et al., 2023), and to identify ethical concerns in ACL papers (Karamolegkou et al., 2025). Our work follows this emerging use of LLMs as tools for analysis, but applies them to the meta-scientific study of research practice itself: we use LLMs to extract structured information on how human annotation is reported across NLP papers.

3 Human Annotation Setup

The annotation unit in this study is an *individual human annotation task* rather than a paper, since papers may contain multiple studies with different reporting levels. A task is defined as a shared annotation setup in which annotators follow the same instructions and annotate the same data. For example, sentiment and stance labeling over the same tweets constitute separate annotation tasks when they use different instructions. We distinguish three intended uses: *resource creation*, such as constructing a new NER corpus; *human performance*, such as establishing a human QA baseline; and *model-output evaluation*, such as rating the fluency of machine translations. In our experience, identifying the number of human tasks within papers was a non-trivial preliminary step. In cases of disagreement, a third annotation was commissioned. Papers with unresolved disagreements after the third annotation, as well as structurally complex papers judged by at least two annotators as containing more than three annotation tasks, were excluded from the final manual dataset.

Annotations were performed using a structured interface with integrated instructions that cover category rationale, definitions, examples, and coding rules. The process began with pilot annotations and extensive group discussions to refine the taxonomy and calibrate annotator interpretations. Closed-list

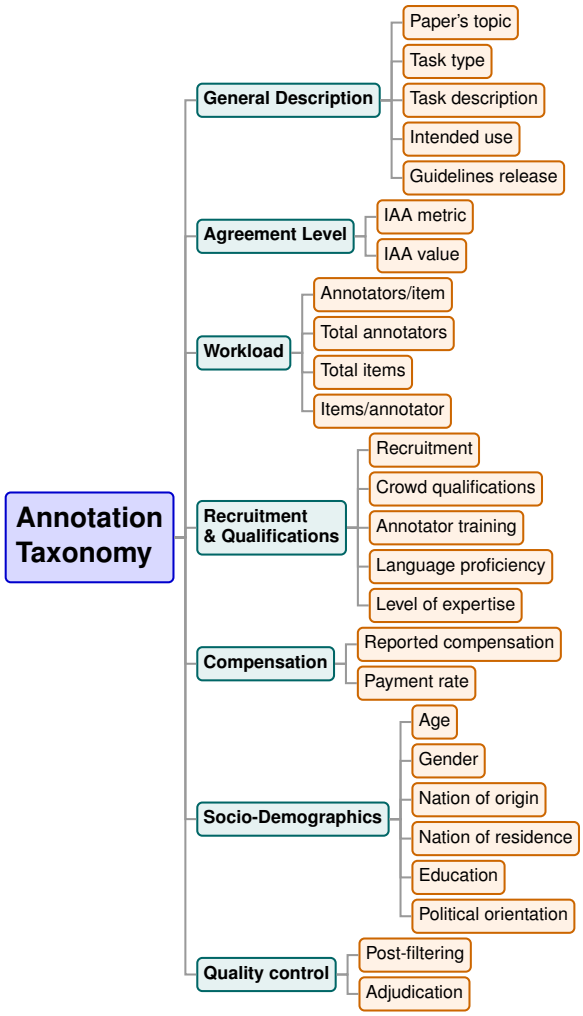


Figure 1: **Task-level taxonomy for annotation-reporting analysis.** The taxonomy groups 26 reporting categories into seven aspects: general task description, agreement, workload, recruitment and qualifications, compensation, socio-demographics, and quality control.

selection was preferred over open-ended extraction to ensure consistent coding and easier aggregation of results. For some categories, annotators relied only on explicit statements in papers, while other tasks required informed judgment. The final taxonomy captures seven aspects of human annotation reporting across 26 categories. The full description of categories, permissible values, and instructions are provided in Appendix D (Table 5). A top-level overview of the taxonomy is shown in Figure 1. This taxonomy provides a unified framework for mapping heterogeneous human annotation tasks in ACL papers. Each paper was annotated by at least two independent annotators, including co-authors of this paper.

In total, 12 annotators contributed to the annotation process: 2 professors, 2 postdoctoral researchers, 6 PhD students, and 2 master's stu-

Subset	Counts			Ratios	
	Input	Annotated	Tasks	Ann. / Input	Tasks / Paper
ANNOTATED$_{\text{GOLD}}$	61	41	72	0.672	1.756
ANNOTATED$_{\text{LLM}}$	1,995	1,603	2,667	0.820	1.664

Table 1: ANNOTATED subsets. Input papers are candidate papers before filtering; annotated papers are papers with annotatable human annotation tasks; annotation tasks are task-level annotation records.

	ANNOTATED$_{\text{LLM}}$	ANNOTATED$_{\text{GOLD}}$
Papers	1,603	41
Tasks	2,667	72
Tasks/paper	1.66 ± 0.82	1.76 ± 0.80
Pages/paper	17.1 ± 6.9	17.0 ± 6.1
Selected topics (% of papers)		
Open-ended generation	44.2	9.8
Semantics/inference	25.1	2.4
Subjective/social	21.5	26.8
Linguistics	4.1	24.4

Table 2: Comparison of ANNOTATED$_{\text{LLM}}$ and ANNOTATED$_{\text{GOLD}}$. The subsets are similar in document length and annotation-task density but differ in topic composition. Topic labels are multi-label; selected topics show the largest or most relevant distributional differences.

dents. All annotators were proficient in reading academic English. Human annotations were finalized through a two-stage adjudication process involving three annotators from the same pool of annotators: two independent annotations of each identified task were compared, disagreements were discussed to reach consensus, and a third annotator adjudicated unresolved cases when necessary. The resulting consensus labels form the gold standard used to evaluate LLM extraction quality; we refer to this subset as ANNOTATED$_{\text{GOLD}}$. Figures 15, 16, and 17 in Appendix F show an annotated paper excerpt and information extracted from it.

4 ANNOTATED Dataset

We began by retrieving papers from ACL Anthology publications appearing between 2018 and 2025 (three years before and after the introduction of the ACL checklist) in the main and Findings proceedings of ACL, EMNLP, NAACL, TACL, EACL, and AACL. Candidate papers were identified by matching 34 curated annotation-related keywords against titles and abstracts, including terms such as *manual annotation* and *human evaluation*. The keyword list (see Appendix A) was iteratively refined during pilot annotation sessions.

ANNOTATED$_{\text{GOLD}}$ is the manually annotated and adjudicated gold-standard set used to evaluate LLM extraction quality. It consists of 41 papers with reliably annotatable content out of 61 initially selected to construct a manual benchmark. The papers were excluded when the number of annotation tasks could not be resolved after a third annotation, or when at least two annotators identified the paper as containing more than three difficult-to-annotate human annotation tasks. The retained papers contain 72 distinct human annotation tasks. The resulting adjudicated dataset is used exclusively for evaluating LLM performance. The size of ANNOTATED$_{\text{GOLD}}$ reflects the cost of expert annotation: the manual annotation and adjudication effort is estimated at approximately $6,300$ Euros in personnel time, with details provided in Appendix C.

ANNOTATED$_{\text{LLM}}$ is the large-scale LLM-extracted set used for the main analysis. The keyword-based sampling yielded 1,995 candidate papers. After filtering papers without annotatable human annotation content, ANNOTATED$_{\text{LLM}}$ contains 1,603 papers and 2,667 extracted annotation tasks. ANNOTATED$_{\text{LLM}}$ strictly excludes all 61 papers sampled for ANNOTATED$_{\text{GOLD}}$, ensuring that the gold-standard evaluation set remains separate from the large-scale analysis. Table 1 summarizes the number of input papers, retained papers, and annotation tasks in each subset.

This venue restriction applies to the large-scale ANNOTATED$_{\text{LLM}}$ analysis. A small number of papers from non-ACL venues were included only during construction of ANNOTATED$_{\text{GOLD}}$ and early taxonomy development, allowing us to test the taxonomy on a broader range of AI and multimodal annotation settings. These papers are not part of the large-scale analysis. We exclude LREC from the present analysis because our temporal study targets the ACL venue ecosystem in which the Responsible NLP Checklist was introduced; LREC has a distinct, strongly resource-oriented research and reporting culture, and is a possible direction for future work.

Table 2 compares ANNOTATED$_{\text{GOLD}}$ with the large-scale ANNOTATED$_{\text{LLM}}$ corpus. The two subsets are highly similar in document length (17.0 vs. 17.1 pages on average) and annotation density (1.76 vs. 1.66 tasks per paper). Their topic composition differs more noticeably (JSD $= 0.161$; Cramér's $V = 0.18$), with ANNOTATED$_{\text{GOLD}}$ containing relatively more linguistics papers and fewer open-ended generation papers.

Sampling validation. ANNOTATED$_{\text{LLM}}$ is constructed through keyword-based retrieval, which may introduce sampling bias relative to all ACL-

venue papers. To assess the effect of this sampling strategy, we compare it with a stratified random sample from the same venue-year scope. Although the comparison reveals statistically significant differences in frequency distributions for some category-value pairs, the effects are generally modest: the average absolute difference across 67 observed value frequencies does not exceed 5.2 percentage points, with effects lacking directional consistency (see detailed analysis in Appendix A.1). We interpret these deviations as sufficiently limited for the purposes of LLM extraction, particularly given the computational, financial, and environmental costs associated with processing the full ACL venue-year population. Accordingly, we treat ANNOTATED$_{\text{LLM}}$ as a high-recall, annotation-focused corpus suitable for studying reporting practices in papers likely to involve human annotation, without claiming it represents all ACL-venue papers.

5 Experiment Setup

Extraction. We evaluate six LLMs under a unified extraction protocol: three proprietary (`Gemini-3.1-Pro` (Gemini Team, 2025), `Gemini-3.1-Flash-Lite` (Gemini Team, 2025), `GPT-4.1` (OpenAI, 2025)) and three open-weight (`Qwen3.6-27B` (Qwen Team, 2026), `gemma-4-31B-it` (Google, 2026), `gpt-oss-120b` (OpenAI, 2025)). We include both proprietary and open-weight models because prior work shows that closed models often provide strong performance, while open models offer advantages in cost and transparency (Oketch et al., 2025). Each model is prompted, with the paper converted to plain text, to populate the same 26-category taxonomy used in human annotation, outputting one JSON record per distinct annotation experiment identified in the paper. The prompt (reproduced in full in Appendix E) encodes the complete taxonomy with exact allowed values, field-level decision rules, interdependency constraints, and a self-audit checklist that instructs the model to continue scanning beyond the first annotation section found: a common failure mode that causes multi-experiment papers to be under-extracted. Model output is constrained to a JSON-defined schema via structured-output APIs where available, eliminating free-form generation and enforcing value types at decoding time. To handle long documents within the context constraints of open-weight models, we apply annotation-section chunking: sections likely to contain annotation metadata (e.g., *Annotation Procedure*, *Human Evaluation*, *Data Collection*) are identified via regex heuristics and passed to the model within an 8,000-token budget; papers with no detectable annotation sections are truncated to 20,000 tokens and processed in full. Proprietary models with million-token context windows receive the full paper text without truncation. Papers for which the model identifies no human annotation experiments are flagged and excluded. The final ANNOTATED$_{\text{LLM}}$ dataset is constructed with Gemini-3.1-Pro, which receives the complete paper text. The chunking and truncation procedures described above apply only to the open-weight models included in the model-comparison experiment.

Evaluation. LLM-extracted values are compared to gold-standard labels at the level of identified tasks. For papers with multiple tasks, we first align each gold task with the corresponding LLM-extracted task using the short task description recorded during annotation. For example, a gold task described as *sentiment labeling of tweets* can be matched to a model-extracted task described as *tweet sentiment annotation*. Before comparison, all values are normalized to a canonical form: IAA metric surface variants (e.g., *Cohen kappa*, κ, *Cohen's K*) are collapsed to a single canonical label per metric; numeric fields accept shared integer tokens as agreement (allowing, e.g., *27.9K* and *27900* to match); list-valued fields (`iaa_metric_name`, `papers_topic`, `intended_use`) are compared via set intersection, with partial overlap counting as agreement. We report per-value and per-category exact-match agreement rates alongside Krippendorff's α.

REPORTAGE SCORE. To assess the level of reporting associated with human annotation in NLP papers, we compute a REPORTAGE SCORE for each identified annotation task. We start from the assumption that all annotation attributes in our inventory are desirable to report when they are applicable to a given annotation setup. The score therefore measures the proportion of applicable annotation attributes that are explicitly reported.

Formally, for an annotation task t, let A_t denote the set of applicable reporting attributes and let $R_t \subseteq A_t$ denote the subset of applicable attributes that are reported. The REPORTAGE SCORE is de-

fined as:

$$\textsc{Reportage Score}(t) = \frac{|R_t|}{|A_t|}.$$

The denominator A_t is task-sensitive because not every reporting attribute is applicable to every annotation setup. Some attributes have no applicability restrictions and are therefore expected for all tasks: *guideline availability, recruitment strategy, annotator training, expertise, language proficiency, education, number of annotators, number of annotated items, compensation, and post-annotation filtering*. Other attributes are excluded whenever the corresponding annotation setup makes them irrelevant. For example, inter-annotator agreement and adjudication are excluded for single-annotator tasks, including text production assignments; crowdworker screening is excluded when crowdsourcing is not used; and redundant workload fields are excluded when they encode information already captured by other categories. For tasks in *Subjective language & social meaning*, five socio-demographic attributes are added to the denominator because annotator background is particularly relevant for interpreting subjective or socially grounded judgments. Thus, failing to report an applicable attribute lowers the score, whereas an attribute that is not applicable to the task does not affect it. Full scoring rules are provided in Appendix B.

6 Results

In §6.1, we first analyze annotation agreement, including both human–human and human–LLM agreement. We then examine broader reporting trends in §6.2, using $\textsc{Annotated}_{\textsc{LLM}}$ and addressing our research questions on annotation intensity, reporting variation across NLP areas, the impact of ACL Responsible NLP Checklist, and the intended use of human judgment.

6.1 Agreement Analysis

Human–Human agreement Overall, human–human exact-match agreement reaches 79.2%, with a macro-averaged Krippendorff's α of 0.585, which is commonly regarded as decent agreement. There is substantial variation across annotation dimensions. Fields corresponding to clearly defined and explicitly reported information, including demographic attributes, compensation, and inter-annotator agreement metrics, consistently achieve high reliability ($\alpha > 0.8$). In contrast, lower agreement is concentrated in categories that require interpretation or inference from less formalized descriptions (e.g., guideline availability, annotator expertise, recruitment, and quality control procedures). This pattern is expected given the nature of the task: disagreement arises primarily in cases where the annotated papers provide incomplete or ambiguous information. Importantly, all annotations were subsequently adjudicated through extensive discussion, yielding a consensus dataset that resolves individual disagreements. Such adjudication can be expected to to produce labels that more closely approximate the underlying ground truth than independent annotations and provides a sufficiently robust benchmark for evaluating LLMs (Klie et al., 2024).

Model Name	Context Len	Agree%	Kripp α
Gemini-3.1-Flash-Lite	1M	72.4	0.50
Gemini-3.1-Pro	1M	**79.9**	**0.61**
GPT-4.1	1M	75.2	0.50
Qwen3.6-27B	262k	63.4	0.42
gemma-4-31B-it	256k	65.8	0.45
gpt-oss-120b	131k	66.9	0.47
Human	–	79.2	0.59

Table 3: Evaluation results comparing various models against human baseline performance.

Human–LLM agreement Table 3 presents the overall performance of our six tested LLMs (three open-source and three proprietary). Gemini-3.1-Pro shows the strongest results. Overall, the best LLM achieves slightly higher macro-level agreement than human–human annotation (79.9% vs. 79.2%) and a marginally higher Krippendorff's α (0.606 vs. 0.585), indicating that model-assisted annotation is not only consistent with human judgments but, in aggregate, slightly more stable against the gold standard. Individual per-category human–human and human–LLM agreement values are reported in Table 6 (Appendix D). Several categories exhibit a mismatch between high observed percentage agreement and moderate or low α, driven by class imbalance and skewed label distributions. Two extreme cases illustrate this effect: for 'political orientation', the absence of positive instances yields perfect agreement but limited interpretability, whereas for 'nation of origin', scarcity of positive cases makes agreement estimates unstable. For rare categories, α can become negative when observed agreement is lower than the agreement expected by chance under the label distribution. Thus, the nega-

tive value for 'nation of origin' in human–human IAA should not be interpreted as substantive disagreement, but as an instability caused by very sparse positive cases. To obtain a symmetric comparison, we additionally compare Gemini-3.1-Pro against the pooled individual pre-adjudication human annotations over 126 matched subtasks. The model reaches 74.8% exact-match agreement and a macro-averaged Krippendorff's $\alpha = 0.602$. Together with the independent human–human baseline of 79.2% agreement and $\alpha = 0.585$, this provides a more directly comparable assessment of model and human annotation consistency.

6.2 Trends in Reporting

After evaluating all six models on ANNOTATED$_{\text{GOLD}}$, we select the best-performing model, Gemini-3.1-Pro, for the large-scale extraction. This model is applied to ANNOTATED$_{\text{LLM}}$ to produce the final dataset used in the downstream analysis, which cost a total of $8,300$ Euros. To analyze temporal trends in methodological reporting and its association with factors such as paper topic, intended use of human judgment, and venue, we use the REPORTAGE SCORE, see §5.

RQ1: Which NLP areas are more human-annotation intensive, and which aspects of annotation practice are more consistently reported? Figure 2 shows reporting rates for individual annotation attributes and their variation across NLP topics. Panel A shows that papers most often report operational details: recruitment (90.4%), annotator expertise (86.5%), and total number of annotated items (86.0%). In contrast, reporting is much weaker for attributes that help readers assess the annotation process itself, including annotator training (18.7%), language proficiency (24.0%), and released annotation guidelines (34.1%). Panel B shows that this imbalance is broadly consistent across NLP areas. Language proficiency is reported somewhat more frequently in papers involving multilingual tasks and linguistic annotation. As expected, annotator demographics are reported somewhat more often in papers dealing with subjective judgments, preferences, and socially constructed meaning (*Subjective language and social meaning* topics), i.e. tasks that investigate social and cultural phenomena through language, such as hate speech, bias, stance, sentiment, and humor annotation. We examine *Subjective language and social meaning* category separately because annotator background, language proficiency, and social positioning can directly affect judgments in these tasks. However, compared with the rest of the dataset, these papers do not show systematically stronger reporting overall (Figure 6). There are some distinctive tendencies: recruitment information and annotator language proficiency are reported slightly more often, crowdworkers and mixed annotator pools are used more frequently, and authors are less often used as annotators (Figures 12 and 7, Appendix F). These results suggest that socially oriented NLP papers are slightly more attentive to annotator sourcing and language background, but still do not consistently report the broader methodological details needed to assess annotation validity.

RQ2: What is the impact of introducing ACL responsible checklist (2022)? The ACL Responsible NLP Checklist (Rolling Review, 2022) asks authors to reflect on data, annotation, and ethical considerations. Accordingly, we test whether its introduction around 2022 coincides with a measurable change in human-annotation reporting. We fit an interrupted time-series regression model that estimates both an immediate post-2022 level shift and a change in the post-2022 trend. Figure 3 shows the fitted interrupted time series regression line with confidence intervals, alongside the observed yearly reportage scores (green dots with error bars) and the counterfactual continuation of the pre-2022 trend. Reportage scores increase steadily between 2018 and 2021, followed by a modest upward level shift in 2022 and a visibly flatter post-2022 trend relative to the counterfactual continuation of the pre-checklist trajectory. Our analysis finds no evidence that the introduction of the ACL checklist in 2022 produced a substantial immediate improvement in reporting scores. Although reportage quality continues to improve after 2022, the rate of improvement appears lower than in the preceding years, suggesting a flattening of the earlier upward trend. We can offer two explanations for this finding. First, aggregate averages may obscure heterogeneous developments across different categories of annotation studies, potentially masking opposing trends in reporting practices. Second, the upward trend already visible before 2022 suggests that the checklist may have formalized practices that were already emerging in the field. Moreover, because the ACL Responsible NLP Checklist (introduced at NAACL 2022) was based on the NeurIPS 2021 checklist, awareness

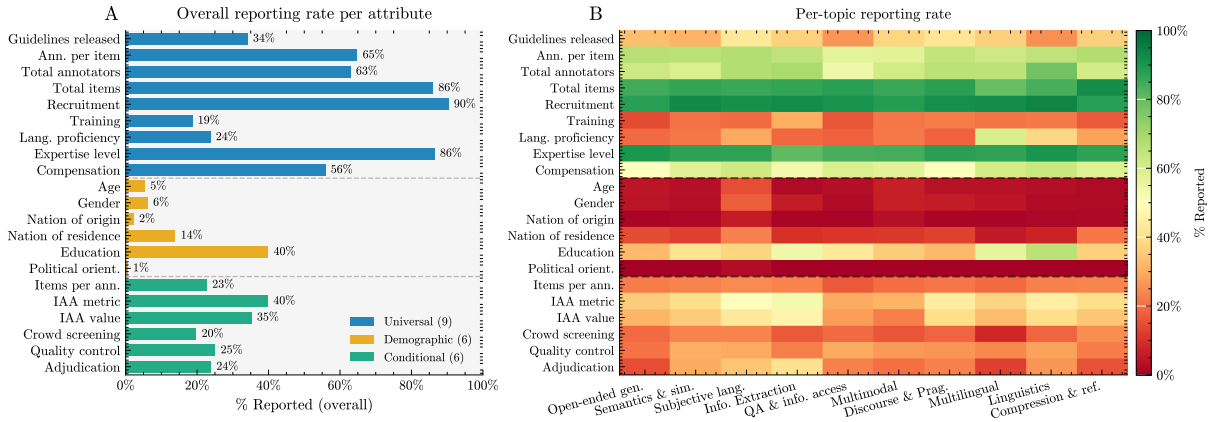


Figure 2: **Reporting patterns across annotation attributes.** Panel A shows the overall percentage of annotation tasks for which each attribute is reported. Panel B shows the corresponding reporting rates by NLP topic. Darker green cells indicate higher reporting rates within a topic; orange and red cells indicate lower reporting rates.

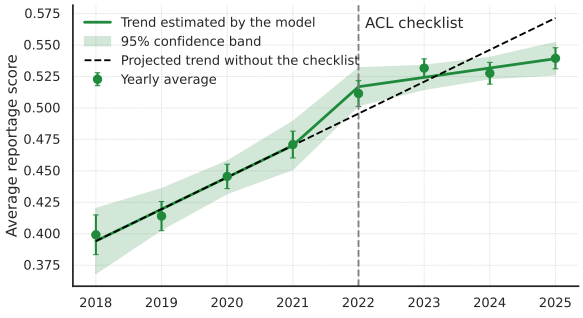


Figure 3: **Interrupted time-series analysis of REPORTAGE SCORE before and after the ACL Responsible NLP Checklist.** Points show observed yearly mean REPORTAGE SCORE values, error bars indicate standard errors, the solid line shows the fitted trend, and the dashed line shows the counterfactual continuation of the pre-2022 trend. The vertical dashed line marks the checklist introduction.

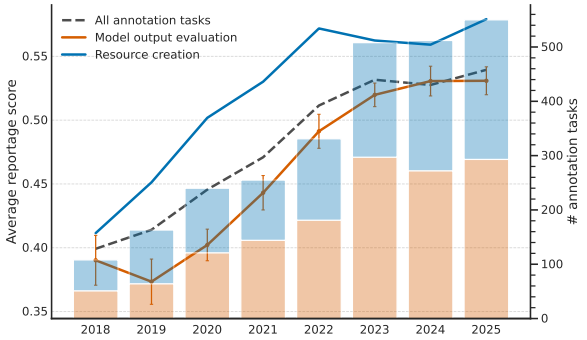


Figure 4: **Mean REPORTAGE SCORE by intended use of human annotation.** Lines show yearly mean REPORTAGE SCORE values for all annotation tasks, model-output evaluation tasks, and resource-creation tasks; error bars indicate standard errors. Stacked bars in the background show the annual number of annotation tasks in the two intended-use categories, illustrating changes in task composition over time.

of annotation reporting standards may have begun increasing earlier, which aligns with the sharper rise in reporting levels observed around 2021 for some types of annotation studies. A venue-specific interrupted time series analysis for ACL, EMNLP, and NAACL and methodology are provided in Appendix F. Table 7 in Appendix F offers a detailed per-category comparison for pre- and post-2022 periods.

RQ3: Does the intended use of human judgment affect reportage quality? We use the *intended use* labels from our taxonomy to compare annotation tasks whose primary purpose is resource creation, model output evaluation, or human-performance benchmarking. These labels are assigned during the human and LLM annotation process evaluated in §6.1. Because human-performance benchmarking is comparatively rare, Figure 4 focuses on the two most frequent uses: resource creation and model output evaluation.

While overall reportage quality improves over time, resource creation studies consistently report substantially more methodological detail than model evaluation studies, with the gap persisting throughout the observation period. In particular, evaluation-focused work more frequently omits information about annotator recruitment, compensation, training, and quality-control procedures. To further examine this pattern, we fit a logistic regression model predicting the reporting of quality-control-related categories in our annotation scheme, including adjudication, post-filtering, annotator training, IAA metrics, and recruitment source (full results in Appendix F). Resource creation studies are substantially more likely to report

such measures ($p < 0.001$), while publication year shows only a weak positive association after controlling for annotation use type. The same overall pattern holds for recruitment strategy and compensation reporting, suggesting that model evaluation studies systematically underreport key aspects of annotation methodology.

The trajectories by intended use provide evidence against a purely compositional explanation of the flatter post-2022 reportage trend observed in RQ2. Such an explanation could arise from the post-2022 growth of model-output evaluation tasks, which tend to report less methodological detail. Despite sharply diverging task volumes after 2022 (stacked bars in Figure 4), reporting scores for both resource-creation and model-output-evaluation tasks flatten at approximately the same time. In particular, the earlier increase in reporting for resource-creation tasks also levels off after 2022, even as their share of the two task types decreases with the growth of model-output evaluation. Thus, the aggregate post-2022 flattening cannot be explained solely by a shift in task composition, although the observational design does not establish a causal effect of the checklist.

7 Conclusion and Recommendations

Human annotation remains a foundation of NLP research, but our results show that the field still lacks consistent standards for documenting who annotators are, how annotation is organized, and how annotation quality is controlled. We provide the first large-scale, task-level audit of human annotation reporting across major NLP venues. Our findings show clear progress, but also persistent blind spots: NLP papers often report who annotators are in broad terms, especially recruitment source, expertise, and annotation scale, but they provide a much less complete picture of whether annotators are trained, qualified, compensated, demographically located, or involved in systematic disagreement resolution. In particular, compensation, socio-demographics, training, adjudication, and agreement values remain substantially underreported. These omissions matter: without such information, readers cannot assess annotator suitability, potential background effects, disagreement handling, or whether the labels support the paper's claims. Our findings may also help explain why human annotations are often difficult to reproduce in NLP (Belz et al., 2023).

We draw three recommendations.

First, papers should **report all relevant annotation details for every human annotation task** they contain. This includes attributes applicable across annotation setups without restrictions: annotator recruitment, number of annotators, number of annotated items, annotators per item, training, language proficiency, expertise, compensation, quality control, and access to annotation guidelines, as well as any additional attributes relevant to the particular task and annotation procedure.

Second, **papers on socially grounded and subjective language tasks should report relevant socio-demographic and background information about annotators**, such as age, gender, nation of residence, nation of origin, and political orientation, as these characteristics may shape the judgments being collected and their interpretation.

Third, **model-evaluation studies require substantially more annotation details**. Our analyses show that when annotation is used for model evaluation, it is documented less than when annotation is used for resource creation, even though such evaluations often support central empirical claims.

These details are not optional metadata; they are necessary for readers to assess whether the annotation procedure is reliable, interpretable, and reproducible.

As LLMs are used as annotators and evaluators, analogous reporting standards are needed for machine annotation, including the exact model and version, prompts and decision rules, context handling, decoding or sampling configuration, and procedures for validation and evidence tracing. Developing such standards is a natural extension of the framework proposed here.

Overall, our study reframes annotation reporting as a measurable methodological practice. The proposed taxonomy and LLM-assisted pipeline provide a scalable basis for monitoring whether NLP papers document human annotation in ways that support reliability and reproducibility.

Limitations

A first limitation concerns the design and reliability of the annotation taxonomy. Some categories require fine-grained interpretation of incomplete or inconsistent paper descriptions, making perfect agreement difficult. We retained these categories because they capture information important for assessing the quality and transparency of the annota-

tion process. The moderate agreement observed for some attributes should therefore be read not only as a limitation of our annotation scheme; it is also evidence that NLP papers often describe annotation procedures in ways that are difficult to interpret consistently. A second limitation is that the large-scale analysis relies on LLM-based extraction. Although we validate the extraction pipeline against a human-adjudicated gold standard, extraction errors remain possible and may affect individual category estimates. Moreover, ANNOTATED$_{\text{GOLD}}$ is necessarily limited in size because task-level expert annotation is costly; the manual annotation and adjudication effort is estimated at approximately 6,300 Euros in personnel time, as detailed in Appendix C. We therefore interpret our results as aggregate reporting patterns rather than definitive judgments about individual papers. Finally, ANNOTATED$_{\text{LLM}}$ is an annotation-focused corpus constructed through keyword retrieval and should not be interpreted as a fully representative sample of all ACL-venue papers.

A third set of limitations concerns the granularity of our taxonomy and released extractions. First, while we record annotator language proficiency and whether a paper concerns multilingual or translation research, we do not represent the specific language identity of the annotated dataset as a structured field. Reporting patterns may differ systematically across languages, particularly in low-resource settings, and future versions of the taxonomy should therefore record target-language identity explicitly. Second, ANNOTATEDLLM does not provide calibrated per-instance confidence scores. We instead report category-level human–human and human–LLM reliability and will release these values in machine-readable form, together with verbatim evidence for selected extraction fields; nevertheless, users should exercise additional caution for low-agreement categories. Finally, because keyword retrieval may favor papers that foreground human annotation, our absolute reporting rates may be optimistic relative to the broader ACL literature and should not be interpreted as population estimates.

Ethics Statement

This work analyzes publicly available scientific papers and does not introduce new experiments involving external human subjects. The manual annotation was conducted by the authors and collaborators on published research articles. Nevertheless, the study raises ethical questions about annotation labor and research transparency. In particular, our findings show that compensation, annotator demographics, training, and quality-control procedures are often underreported in NLP papers. We do not interpret missing reporting as evidence that these practices were absent; rather, we treat it as a transparency gap that limits readers' ability to assess annotation validity, fairness, and reproducibility. Our LLM-assisted extraction pipeline also introduces ethical considerations. Automated extraction may misclassify individual papers or categories, especially when papers describe annotation procedures ambiguously. For this reason, we validate the pipeline against a human-adjudicated gold standard and report aggregate trends rather than using the system to rank or criticize individual papers. Released dataset and code should be used to study community-level reporting practices and to support better documentation standards, not to assign blame to individual authors.

Broader Impact

This work aims to improve the transparency and reproducibility of human annotation in NLP. By providing a task-level taxonomy, a reporting-completeness metric, and an LLM-assisted extraction pipeline, the study offers tools for monitoring how the field documents annotation labor, quality control, and annotator characteristics. The most direct positive impact is methodological: clearer reporting can help reviewers, readers, and future researchers assess whether annotation-based claims are reliable, comparable, and reproducible. It can also support future checklist design by identifying which annotation details are most often missing. Importantly, reporting requirements may encourage researchers to reflect more carefully on annotation procedures before data collection begins. At the same time, reporting standards must be applied carefully. Calls for more demographic information should not pressure authors to collect sensitive personal data when it is unnecessary, unsafe, or ethically inappropriate. Instead, task-sensitive reporting should mean documenting information that is relevant to interpreting annotation outcomes, while respecting annotator privacy and consent. More broadly, our goal is to promote more deliberate consideration and transparent reporting of the attributes relevant to each annotation setup.

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A Datasets

The following 34 keywords were used to identify papers involving human annotation based on matches in titles and abstracts.

- Amazon Mechanical Turk
- annotator
- crowdworker
- crowdsourcer
- crowdsourcing
- expert annotation
- expert evaluation
- expert opinion
- HIL
- human annotation
- human annotator
- human benchmark
- human dataset
- human evaluation
- human expert
- human in the loop
- human judgment
- human judgement
- human performance
- human perception
- human rating
- human raters
- human respondent
- human study
- human-in-the-loop
- labeler
- manual annotation
- manual evaluation
- manual rating
- manually annotated
- manually evaluated
- MTurk
- Prolific
- student worker

A.1 Random-sample validation of keyword retrieval

The large-scale extraction set, $\text{ANNOTATED}_{\text{LLM}}$, is constructed using keyword-based retrieval over paper titles and abstracts. This strategy is designed to increase the concentration of papers likely to contain human annotation tasks, but it may also change the composition of the resulting sample. To assess this effect, we compare $\text{ANNOTATED}_{\text{LLM}}$ with a stratified random sample drawn from the same venue-year scope.

The random sample contains 3,000 papers drawn across the same venues and publication years. Of these, 1,085 papers contain annotatable human annotation content, corresponding to a retention rate of 36%. In contrast, keyword-based retrieval yields 1,603 retained papers out of 1,995 candidates, corresponding to a retention rate of 82%. This confirms that keyword retrieval substantially improves the efficiency of identifying papers with human annotation content.

To check whether our sampling strategy introduced any bias, we compare the distributions of taxonomy-value pairs between the keyword-retrieved set and the random sample. For each category-value pair, we compute the normalized frequency in both samples and test the difference in proportions using a χ^2 test. Figure 5 shows only category-value pairs with statistically significant differences at $p < 0.05$. To avoid duplicate visualizations for binary variables, the figure reports only the positive or reported value where applicable.

The comparison shows that keyword retrieval introduces some statistically significant composition differences. However, these differences are generally modest: among statistically significant category-value pairs, the average absolute difference is 5.2 percentage points. Only a small num-

ber of category-value pairs exhibit substantial deviations, with extreme cases reaching -16 and $+18$ percentage points in opposite directions. In other words, the keyword-retrieved set is not simply higher or lower on all reporting dimensions. Instead, it over-represents some category values and under-represents others.

We therefore interpret ANNOTATED$_{\text{LLM}}$ as an annotation-focused corpus rather than a fully representative sample of all ACL-venue papers. The random-sample comparison suggests that keyword retrieval changes sample composition, but does not reveal a single systematic directional bias across reporting categories. This supports our use of ANNOTATED$_{\text{LLM}}$ for analyzing reporting patterns among papers likely to involve human annotation, while motivating caution in generalizing the exact proportions to all ACL Anthology papers containing human annotation tasks.

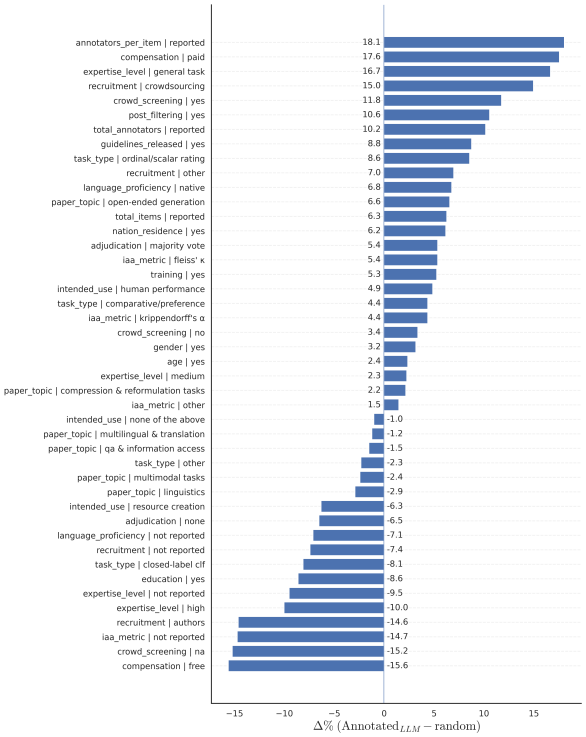


Figure 5: Significant proportional differences ($\Delta\%$) between filtered and random samples across category–value pairs (chi-square test, $p < 0.05$). Bars are sorted by effect size and centered at zero to indicate direction (over- vs under-representation in the keyword-based sample). For binary and binarized string fields, only the significant proportional deltas for the positive value (e.g., "yes", "reported") are shown to avoid duplication.

B REPORTAGE SCORE Calculation

The REPORTAGE SCORE is computed as the proportion of reported annotation attributes relative to the set of attributes applicable to a given annotation task. The full annotation schema contains 21 reporting categories grouped into:

- 10 universal attributes expected for all studies (guidelines release, recruitment strategy, training procedures, level of expertise, language proficiency, educational level, number of annotators, number of annotated items, compensation, post-annotation filtering)
- 5 socio-demographic attributes describing annotator background (age, gender, nation of residence, nation of origin, political orientation), which are expected in tasks annotating subjective or socially-constructed language phenomena
- 6 attributes that can be excluded as inapplicable or redundant given other types of information reported in the paper for the task (inter-annotator metric and value, annotator per item, items per annotator, adjudication, crowdworker screening)

The denominator is adjusted dynamically to avoid penalizing papers for omitting inapplicable or redundant information, following a set of rules as follows. Specifically, the papers belonging to *Subjective language & social meaning*, the denominator includes the five socio-demographic characteristics, for other papers, they are viewed as optional. Inter-annotator agreement (IAA) and adjudication are excluded from the denominator for single-annotator studies (i.e., when `total_annotators = 1` or `annotators_per_item = 1`, which is often the case for text production tasks); the IAA value is excluded if no IAA metric is reported; crowdworker screening is excluded when recruitment strategy does not involve crowdsourcing; and adjudication is excluded for tasks without open-ended annotation.

To avoid double-counting, redundant fields are excluded when they encode equivalent information. For example, if `total_annotators` and `annotators_per_item` have the same number, one of them is omitted. The `items_per_annotator` is excluded when recoverable from other reported fields. Missing information is treated as incomplete reporting;

for instance, reporting an IAA metric without the corresponding value does not count as valid reportage. Fine-grained compensation extensions such as payment rate are treated as refinements rather than independent reporting dimensions and are excluded from scoring.

The final score is computed separately for each annotation task in a paper using the resulting set of applicable reporting categories.

C Annotator demographics

In total, 12 annotators contributed to the annotation process: 2 professors, 2 postdoctoral researchers, 6 PhD students, and 2 master's students. Among the annotators, 5 identified as women and 7 as men. The annotator pool included individuals originating from Italy, Germany, India, China, and Russia. All annotators reported fluent English proficiency, as all annotated papers were written in English. At the time of annotation, most annotators were affiliated with institutions located in Germany and Austria. Annotation was carried out on a part-time basis alongside regular academic activities. Professors contributed approximately 10 hours in total, while postdoctoral researchers, PhD students, and master's students each contributed approximately 50 hours overall. Based on the 2026 standardized academic personnel rates published by the Deutsche Forschungsgemeinschaft, the estimated total personnel cost of the annotation process was approximately 6,300 Euros.

D Details on Human Annotation

Paper ID	Tasks	Domain	Year	Venue
Wang et al. (2020)	3	Task-Oriented Dialogue Learning	2020	ACL
Jin et al. (2020)	3	Headlines Generation with Controlled Styles	2020	ACL
Zeinert et al. (2021)	1	Misogyny detection	2021	ACL
Zhang et al. (2021)	3	Text Summarization	2021	ACL
Jiang and Riloff (2021)	1	Prototypical function labeling of physical artifacts	2021	ACL
Asai and Choi (2021)	2	Information-Seeking QA	2021	ACL
Haber et al. (2023)	3	Detection of Multilingual Online Attacks	2023	ACL
Sai B et al. (2023)	1	Meta-Evaluation of Machine Translation Metrics	2023	ACL
Agnew et al. (2023)	2	Poetry Generation	2023	ACL
Xiong et al. (2023)	2	Crowd-Annotated Named Entity Recognition	2023	ACL
El Kheir et al. (2024)	2	Dialectal Sound and Vowelization Recovery	2024	ACL
Ide et al. (2025)	1	MWE Corpus Creation	2025	ACL
Zou (2025)	1	Poetry Generation	2025	ACL
Loginova and Loguinova (2025)	2	Deep Temporal Reasoning in Video Language Models	2025	ACL
Breit et al. (2021)	1	Target Sense Verification	2021	EACL
Patro et al. (2018)	1	Visual Question Generation	2018	EMNLP
Weller and Seppi (2019)	1	Humor Detection	2019	EMNLP
Jiang et al. (2020)	2	Crowdsourced Semantic Role Labeling	2020	EMNLP
Zeng et al. (2020)	1	Large-scale Medical Dialogue Datasets	2020	EMNLP
Chakrabarty et al. (2021)	1	Neural Poetry Translation	2021	EMNLP
Mousi et al. (2023)	3	Concept-Based Model Interpretability	2021	EMNLP
Lin et al. (2023)	3	Counter-argument Generation	2023	EMNLP
Zelikman et al. (2023)	3	Generating and Evaluating Educational Tests	2023	EMNLP
Doneva et al. (2024)	1	Biomedical NER Corpus Creation	2024	EMNLP
Kostikova et al. (2024)	1	Solidarity Detection	2024	EMNLP
Walsh et al. (2024)	1	Poetry Evaluation	2024	EMNLP
Altemeyer et al. (2025)	2	Argument Summarization and Evaluation	2025	EMNLP
Mathur et al. (2025)	3	Social Reasoning	2025	EMNLP
Ivetta et al. (2025)	1	Community-Based Bias Evaluation	2025	EMNLP
tse Huang et al. (2025)	1	Social Biases in Vision Language Models	2025	EMNLP
Chen et al. (2025a)	2	Benchmarking for Translating Classical Chinese Poetry	2025	EMNLP
Belouadi et al. (2025)	1	Text-to-Image Generation	2025	ICCV
Greisinger and Eger (2026)	3	Evaluation of TikZ Generated Code	2026	ICLR
Khot et al. (2021)	1	Question Answering	2021	NAACL
Zhao et al. (2024)	1	Gender Bias in Chemical NER Models	2024	NAACL
Li et al. (2024)	2	Contradiction Detection	2024	NAACL
Kang et al. (2024)	2	Synthetic Text Data Inspection	2024	NAACL
Hengle et al. (2024)	2	Counterspeech Generation	2024	NAACL
Vasilev et al. (2025)	2	Text-to-Image Generation	2025	NAACL
Belouadi et al. (2024)	1	Text-to-Image Generation	2024	NIPS
He et al. (2024)	2	Human-Like Translation Strategy	2024	TACL

Table 4: Double-annotated and adjudicated papers included in the gold dataset.

Aspect	Category	Values	Instruction
General Description	Paper's topic		This category captures the core problem the paper introduces or solves, as stated in the abstract or introduction. A paper gets exactly one label, determined by the dominant topical focus of contribution, usually found in abstract. The 10 drop-down options are adopted from ACL categories. In case of hybrid papers, decide what is their primary focus and contribution. If in doubt, assign at most two categories.
		Linguistics	Core linguistic representation, word to sentence levels (e.g. PoS, parsing, NER, semantic roles, word sense disambiguation, entity linking).
		Discourse/Pragmatics	The study of cross-sentence meaning and contextual semantics, relations between sentences/clauses in a text and text-level linguistic phenomena (coreference resolution, discourse relations, implicature/ellipsis resolution, topic segmentation, dialogue act tagging without generation).
		Information Extraction	Advances in text to knowledge bases or structured schema conversion (knowledge graph construction, slot filling, relation/event/temporal prediction).
		QA/Information access	Answering questions from text or corpora (extractive, open-domain, multi-hop QA, both deterministic SQuAD and open-ended).
		Semantics, inference & similarity	Judging meaning relations between texts (entailment, reasoning benchmarks, paraphrase-detection, STS, NLI).
		Subjective language & social meaning	Topics that explore social and cultural phenomena through language (sentiment, emotion, stance, humor, sarcasm, toxicity, bias, propaganda, authorship attribution).
		Open-ended generation	Open-ended output production (dialogue/story/poetry/jokes/argument generation from a given topic or context, instruction following).
		Rewriting tasks	Unlike generic generation, this task transforms a given text and is required to remain faithful to the input in some aspect (summarisation, simplification, paraphrase-generation, style transfer, rewriting under constraints).
		Multilingual/Translation	Cross-linguistic mapping (MT, cross-lingual transfer evaluation, code-switching, multilingual generation).
		Multimodal tasks	Language from other modalities (image/graphics captioning, visual QA, video description, multimodal assistants).
	Task type		Select the type of task performed by human annotators.
		Closed-label classification	Tasks where annotators choose from predefined categories (binary, multi-class, multi-label, tagging with a fixed schema).
		Ordinal/scalar rating	Tasks where annotators pass judgments on a scale (Likert scale, 1-5 ratings, graded acceptability, quality scores).

Aspect	Category	Values	Instruction
		Span-level or structural annotation	Annotators mark parts of the input or define structure (NER spans, coreference links, constituency/dependency trees, alignment).
		Comparative/preference judgments	Relative evaluation of two or more items (A/B choice, ranking, pairwise preference, best–worst scaling).
		Open-ended generation or transformation	Annotators produce new text or modify input (summarization, simplification, paraphrasing, translation, explanation writing).
		Other	None of the above
	Task identity	[User input]	Record the specific name of each human annotation subtask as reported in the paper. This allows matching the order of subtasks within a paper. Especially important for papers reporting up to three annotation studies, but apply to all papers. How to extract: Use the exact name or description of the subtask as given in the paper. Do not create new names or abbreviations.
	Intended use		Categorizes types of human annotation studies by their purpose. Select Resource creation and Human performance when an annotated dataset is used as a human baseline, without collecting additional judgments.
		Resource creation	Human judgment are used to build a dataset, a gold standard, a benchmark or a train/test set for potential further use, including for a subsequent evaluation of their own models in the same study.
		Human performance	Humans solve the same task as the machine to provide the human upper-bound and to compare human-machine performances.
		Model output evaluation	Collecting human judgment to evaluate automatically generated items.
		None of the above	A fallback option
	Guidelines release		Indicates whether the annotation guidelines released by the paper allow the experiment to be reproduced.
		Yes	Guidelines are accessible (in the paper, appendix, via a working link or citation of another paper) and contain all information necessary to reproduce the annotation. Select Yes only if someone could realistically repeat the annotation using the provided materials.
		No	Guidelines are not accessible, or do not contain enough information to reproduce the annotation.
Agreement Level	IAA metric	Fleiss' κ, Cohen's κ, Krippendorff's α, F1 agreement, Pearson, Spearman, Majority agreement, Kendall's τ, Total agreement, Other, Not reported	Multiple selection dropdown. If more than one metric is reported, select several items in the order of appearance in the paper. The corresponding values should be entered in the IAA Value column in the same order separated by a pipe (|). Other: some other IAA measure/method is explicitely reported. Not reported: either not applicable (one annotator) or not given in the paper.

Aspect	Category	Values	Instruction
	IAA value	[User input]	A string field. Refrain from inferring values. Separate multiple values (aligned with IAA metric names) with a pipe (|). Enter a string if ranges are reported, or any other form of report is made (not one value). NA: if Other metric is reported or if the value is not directly reported and needs to be inferred. Not reported: either not applicable (one annotator) or not given in the paper.
Workload	Annotators/item	[User input]	
	Total annotators	[User input]	A number or "Not reported" string. Type "Not reported" if this information is not given.
	Total items	[User input]	
	Items/annotator	[User input]	
Recruitment and Qualifications	Recruitment		This category captures who performed the annotation, i.e., the source from which annotators were recruited.
		Crowdsourcing	Annotators recruited via a crowdsourcing platform or described as crowdworkers.
		Authors	One or more authors performed the annotation.
		Other	Annotators are a specified non-crowd group (e.g., students, experts, hired assistants, colleagues).
		Mixed	Annotators include people from different specified groups.
		Not reported	The source of annotators is not specified as a specific group.
	Crowd qualifications		Applies only to crowdsourced annotators. If the paper does not use crowdworkers, select NA.
		Yes	The paper reports preliminary screening or filtering of crowdworkers to ensure annotation quality (beyond basic socio-demographics).
		No	The paper does not provide any information on quality-related pre-selection or screening of crowdworkers beyond socio-demographics. Choose No if Recruitment is Not reported.
		NA	There is an explicit mention of other-than-crowd sources of annotators (other and authors).
	Annotator training		Is any preliminary annotator training reported (e.g. calibration sessions, trial studies, hands-on prior experience with the annotation tool).
		Yes	Some form of annotator training is reported.
		No	Either not discussed or explicit mention of no training provided.
	Language proficiency		Capture the reported language proficiency of annotators with respect to the language of the annotation task.
		Native	The annotators are directly described as native speakers. If annotators' ethnicity is reported, and it aligns with the language of the annotation task, annotators should be considered native speakers.

Aspect	Category	Values	Instruction
		Non-native	The authors report some level of proficiency in the language of the annotated data (fluent, B2, C1), but not native.
		Mixed	Both native and non-native speakers were employed.
		Not reported	Language proficiency is not directly reported.
	Level of expertise		Indicates the actual expertise of the employed annotators for tasks where specialized knowledge is relevant.
		High	Annotators have specialized domain knowledge relevant to the task (e.g., medical, legal, technical, linguistic experts).
		Medium	Annotators have some relevant background, but are not deeply specialized (e.g., students in a related field).
		Mixed	Reported levels of expertise vary (e.g. experts and students, or students and unfiltered crowd workers).
		Not reported	The task could require specialized knowledge, but the paper does not specify the annotators' expertise.
		General language task	The task does not require specialized knowledge.
Compensation	Reported compensation		Indicates whether annotator compensation is described.
		Paid	Some compensation is mentioned, including "annotators were paid", "full-time employees".
		Free	Cases, where voluntary effort is reported (authors of the paper or colleagues).
		Not reported	No mention of compensation.
	Payment rate		Provide specifics on how compensation is reported for papers where compensation is Paid in Reported compensation.
		Specific numeric rate	A concrete payment is given (e.g., hourly rate, per-task payment, total amount).
		General mention	Compensation is mentioned, but no exact amount is provided (e.g., "annotators were paid", "full-time employees").
		NA	Select NA for cases where the compensation is "Free" or "Not reported" in Reported compensation.
Socio-Demographics	Age	Yes; No	Is age reported?
	Gender	Yes; No	Is gender of annotators reported?
	Nation origin	Yes; No	Is nationality of annotators discussed? Nationality is interpreted as the country of birth.
	Nation residence	Yes; No	Is the nation of residence at the time of annotation discussed? It can be different from the nation of origin.
	Education	Yes; No	Is educational level or field of study reported?
	Political orientation	Yes; No	Is political orientation discussed?

Aspect	Category	Values	Instruction
Quality Control	Post-filtering		Does the paper report any quality control or filtering of annotations after collection?
		Yes	Some form of control and post-annotation filtering is reported or they explicitly report disqualifying experts due to lack of guidelines understanding or removing items due to issues revealed during annotation.
		No	Nothing is reported about filtering the annotation results.
	Adjudication		Captures how annotation disagreements are handled to produce final labels. This refers to explicit procedures for resolving conflicting judgments at the item level (not simply aggregating scores across all annotated items).
		Majority vote	The most common label among annotators is selected.
		Expert adjudication	A designated expert or third-party judge reviews disagreements and assigns the final label.
		Third annotator	Another annotator from the annotator pull steps in for arbitration between diverging opinions.
		Consensus discussion	Annotators discuss and agree on a final label.
		Soft labels	Disagreement is preserved as a label distribution rather than resolved.
		Weighted voting	Votes are weighted based on annotator reliability or expertise.
		Other / mixed	Any combination or alternative method.
		None	No explicit disagreement resolution; annotator outputs are used as-is (e.g., simple averaging or single annotator per item).

Table 5: Annotation categories, values and instructions by aspect.

Table 6: Inter-annotator agreement by category and aspect. Human–Human (N=71) vs Human–LLM (N=64).

Aspect	**Category**	**Human–Human**		**Human–LLM**	
		Agree%	α	Agree%	α
General Description	Paper's topic	77.5	0.677	70.3	0.290
	Task type	73.2	0.660	71.9	0.638
	Intended use	81.7	0.535	79.7	0.592
	Guidelines released	66.2	0.328	71.9	0.433
	Avg	**74.6**	**0.550**	**73.4**	**0.488**
Agreement Level	IAA metric	88.7	0.815	84.4	0.735
	IAA value	93.0	0.897	75.0	0.603
	Avg	**90.8**	**0.856**	**79.7**	**0.669**
Workload	Annotators/item	67.6	0.541	68.8	0.559
	Total annotators	73.2	0.693	71.9	0.681
	Total items	49.3	0.453	57.8	0.546
	Items/annotator	69.0	0.246	76.6	0.253
	Avg	**64.8**	**0.483**	**68.8**	**0.510**
Recruitment & Qualif.	Recruitment	66.2	0.518	59.4	0.384
	Crowd screening	87.3	0.671	87.5	0.586
	Training	83.1	0.625	87.5	0.693
	Language proficiency	77.5	0.549	84.4	0.641
	Expertise level	52.1	0.353	53.1	0.399
	Avg	**73.2**	**0.543**	**74.4**	**0.540**
Compensation	Compensation	84.5	0.741	81.2	0.681
	Payment rate	95.8	0.909	90.6	0.784
	Avg	**90.1**	**0.825**	**85.9**	**0.733**
Demographics	Age	98.6	0.949	96.9	0.858
	Gender	97.2	0.893	96.9	0.872
	Nation origin	88.7	-0.052	96.9	0.653
	Nation residence	84.5	0.336	89.1	0.529
	Education	79.7	0.580	79.7	0.597
	Political orient.	100.0	1.000	100.0	1.000
	Avg	**91.3**	**0.618**	**93.2**	**0.751**
Quality Control	Post-filtering	66.2	0.194	85.9	0.602
	Adjudication	78.9	0.520	79.7	0.539
	Avg	**72.5**	**0.357**	**82.8**	**0.570**
Overall	All categories (macro avg)	**79.2**	**0.585**	**79.9**	**0.606**

E LLM Extraction Prompt

The following prompt was used to guide the Large Language Model in extracting structured data from the scientific papers according to our defined taxonomy.

```
You are extracting structured data about HUMAN ANNOTATION
      experiments from a scientific paper.
Your task is to fill in a rigid taxonomy. You MUST use ONLY
      the exact allowed values specified below.
Do NOT paraphrase, do NOT add explanations, do NOT use
      synonyms.

## Annotation Taxonomy -- EXACT Allowed Values

You MUST output values EXACTLY as listed below -- no
      paraphrasing, no synonyms, no extra words.

### 1. paper_experiment_id
Format: {filename}-1, {filename}-2, etc.

### 2. paper_topic -- LIST of one or more, each EXACTLY one
      of:
  * "Linguistics"
  * "Discourse & Pragmatics"
  * "Information Extraction"
  * "QA & information access"
  * "Semantics, inference & similarity"
  * "Subjective language & social meaning"
  * "Open-ended generation"
  * "Compression & reformulation tasks"
  * "Multilingual & translation"
  * "Multimodal tasks"
  Classify the paper's topic(s) based on its main
      contributions. Use a list with one or more values.

### 3. human_annotation_type -- PICK ONE exactly:
  * "Closed-label classification"
  * "Ordinal / scalar rating"
  * "Span-level or structural annotation"
  * "Comparative / preference judgments"
  * "Open-ended generation or transformation"
  * "Other"
Decision rules:
  - Binary / multi-class / multi-label / tagging with fixed
      schema -> "Closed-label classification"
  - Likert scales, 1-5 ratings, quality scores, graded
      judgments -> "Ordinal / scalar rating"
  - NER spans, coreference, constituency/dependency trees,
      sequence labeling, alignment -> "Span-level or
      structural annotation"
  - A/B choice, ranking, pairwise preference, best-worst
      scaling -> "Comparative / preference judgments"
  - Summarization, paraphrasing, translation, simplification,
      free-text generation -> "Open-ended generation or
      transformation"
  - Anything that does not fit the above five -> "Other"

### 4. subtask_name
Short descriptive name of the HUMAN ANNOTATION subtask --
      what the annotators were asked to label, rate, or judge.
If one experiment covers multiple subtasks that share the
      same annotator pool and annotation round, combine them
      with " | " separator.
  [!] Name the ANNOTATION task, NOT the downstream NLP
      modelling task:
      * Write "hate speech annotation", NOT "hate speech
      classifier"
      * Write "stance labeling", NOT "stance detection model"
      * Write "named entity tagging", NOT "NER model training"
  The name must reflect what humans were asked to DO, not the
      downstream prediction objective.

  [!] Be SPECIFIC -- mirror the paper's own wording, not a
      generic category:
      CORRECT (specific):   "binary online attack
      classification"
      WRONG (too generic):  "attack annotation"
      CORRECT (specific):   "clinical named entity
      recognition (NER)"
      WRONG (too generic):  "NER annotation"
      CORRECT (specific):   "Pairwise Comparison of
      Model-Generated Summaries"
      WRONG (too generic):  "evaluation"
      CORRECT (specific):   "rating arguments on a Likert
      scale across 5 dimensions"
      WRONG (too generic):  "argument rating"
      CORRECT (specific):   "evaluate generated headlines for
      relevance, attractiveness and language fluency"
      WRONG (too generic):  "headline evaluation"

  [!] Use " | " ONLY for subtasks sharing the SAME annotator
      pool and annotation round.
     If tasks differ in annotator group, annotation unit,
      instructions, or objective -> create SEPARATE
      experiments.
     Example of correct " | " use: "Misogyny detection -
      Abusive language detection | Misogyny detection -
      Target identification | Misogyny detection - Hate
      speech categorization"
       (all three done by the same annotators in the same
      study)

  [!] Papers routinely contain 2-3 distinct annotation
      experiments. Common patterns:
      * Different label spaces on the same data (e.g. hate
      speech -> then target type -> then severity)
      * Same task repeated by two different annotator groups
      (e.g. expert gold-standard AND crowd annotation)
      * A text creation task PLUS a separate
      rating/evaluation task on the model output
      * Multiple rounds: pilot annotation + main annotation
      with different guidelines or annotators, if IAA is
      reported separately
      Each of these MUST be a separate experiment with its
      own row.

### 5. intended_use -- LIST of one or more, each EXACTLY one
      of:
  * "Resource creation"    (the annotation is primarily for
      building a dataset/resource)
  * "Human performance"    (the annotation measures human
      performance on a task)
  * "Model output evaluation" (the annotators evaluate
      model-generated outputs)
  * "None of the above"    (does not fit any of the above
      categories)
  Use a list with one or more values. Multiple uses are
      common (e.g. ["Resource creation", "Model output
      evaluation"]).
  - DEFAULT: ["None of the above"]

### 6. guidelines_released -- "Yes" or "No"
  - "Yes" IF any of the following are true:
      * The paper includes a dedicated appendix section with
      detailed annotation instructions (not just a brief task
      description in the main text)
      * A URL / hyperlink to guidelines, an annotation
      manual, or a codebook is provided (even if the link may
      now be broken)
      * The paper cites another paper whose guidelines were
      re-used and those guidelines are thus accessible via
      that citation
      * Supplementary material, a shared GitHub/OSF
      repository, or a dataset release specifically includes
      the annotation guidelines or schema
      * The paper explicitly states that guidelines were
      released or made available
  - "No" if guidelines are only described informally in the
      main text without a dedicated document, appendix, cited
      source, or link; OR if no guidelines information is
      given
  - DEFAULT: "No"
  [!] Common mistake: Do NOT mark "Yes" just because the
      paper describes the annotation task in a paragraph.
      There must be an accessible, standalone guidelines
      document or appendix section, an annotator interface
      screenshot, allowing reproduction.

### 7. iaa_metric_name -- LIST of one or more, each EXACTLY
      one of:
  * "Fleiss' kappa"
  * "Cohen's kappa"
  * "Krippendorff's alpha"
  * "F1 agreement"
  * "Pearson"
  * "Spearman"
  * "Majority agreement"
  * "Kendall's tau"
  * "Total agreement"     (also called percent agreement,
      percentage agreement, observed agreement, raw
```

```
      agreement, exact agreement)
  * "Other"               (any explicit metric not in the
      list above, e.g. ICC, Scott's pi, Gwet's AC1, BLEU,
      Kappa)
  * "Not reported"        (either not applicable because
      there is only one annotator, or no IAA metric is given
      in the paper)
IMPORTANT: If a metric is stated but you cannot identify
      which one, use "Other".
If no metric is mentioned, use "Not reported".
If multiple metrics are reported, list them in the order they
      appear in the paper.
[!] INTERDEPENDENCY: If annotators_per_item == "1" OR the
      task is open-ended free-text generation -> force ["Not
      reported"].

### 8. iaa_value -- LIST aligned 1-to-1 with iaa_metric_name
  - Numeric values as strings (e.g. "0.58", "85.3%", "0.09 to
      0.23") for any real metric, or "NA" if the paper does
      not directly state the value. Refrain from inferring
      values.
  - If ranges are reported or any non-single-value form is
      used, enter as a string (e.g. "0.09 to 0.23").
  - "NA" if iaa_metric_name[i] == "Other" (regardless of
      whether a number is given)
  - "Not reported" if iaa_metric_name[i] == "Not reported"

  - Keep the same number of elements as iaa_metric_name
[!] INTERDEPENDENCY: iaa_value[i] is FULLY DETERMINED by
      iaa_metric_name[i] -- see Field Interdependencies rule
      A above.
[!] IMPORTANT: If you identified an IAA metric, ACTIVELY
      search for its numeric value in: (a) the main text near
      where the metric is mentioned, (b) ALL tables
      (especially agreement/statistics tables), (c) the
      appendix. IAA values are very often reported in tables
      rather than prose. Only output "NA" after checking
      tables.

### 9-12. Workload fields -- integer string or "Not reported"
  - **annotators_per_item**: annotators who labeled EACH
      item. Integer or "Not reported". Ranges like "2-3" are
      acceptable.
  - **total_annotators**: total number of distinct
      annotators. Integer or "Not reported".
  - **total_annotated_items**: total number of
      items/instances annotated IN THIS SPECIFIC EXPERIMENT.
      Integer or "Not reported". If given with K suffix (e.g.
      "27.9K"), keep as-is. If given with commas keep as-is
      (e.g. "2,864").
    [!] Extract only the item count for the annotation
      experiment described. An annotated item is a sample
      under evaluation, not the number of annotated spans or
      the total tokens/lines across annotated items (e.g. the
      number of poems, sentences, documents), NOT the total
      corpus/dataset size that may be larger. If the paper
      provides multiple numbers (e.g. a larger pre-filtered
      pool and a smaller annotated subset), use the smaller
      annotated subset count.
  - **items_per_annotator**: how many items each annotator
      processed. Integer or "Not reported".
  - DEFAULT: "Not reported" -- do NOT guess or calculate from
      other fields.

### 13. human_evaluated_sample -- free-text string
  - Describe the size/scope of the human-evaluated sample
      (e.g. "100", "all", "200 randomly chosen tasks from the
      training data", "500", "random 50")
  - If the paper describes a specific number of items
      evaluated by humans, report that number or description
  - report cross-annotated sample, if different to the total
      number of annotated items  - "Not reported" if no
      human-evaluated sample is described or the size is not
      specified
  - "all" if all items were human-evaluated, especially if
      the intended use of the human labour is to create a
      resource
  - DEFAULT: "Not reported"

### 14. recruitment -- PICK ONE exactly:
  * "Crowdsourcing"  (annotators recruited via a
      crowdsourcing platform or described as crowdworkers)
  * "Authors"        (one or more authors performed the
      annotation)
  * "Other"          (annotators are a specified non-crowd
      group, e.g. students, experts, hired assistants,
      colleagues)
  * "Mixed"          (annotators include people from
      different specified groups)
  * "Not reported"   (the source of annotators is not
      specified as a specific group)
  - DEFAULT: "Not reported"

### 15. crowd_qualifications_screening -- "Yes", "No", or "NA"
  - "Yes" if any pre-screening, qualification test, or
      performance record is described for crowdworkers,
      typically: 95% acceptance rate in previous jobs (not
      socio-demographic filter)
  - "No" if crowdsourcing was used but no screening mentioned
  - "NA" if NOT a crowdsourcing study
  - DEFAULT: "NO" when not crowdsourcing; "No" when
      crowdsourcing
[!] INTERDEPENDENCY (set before checking paper content):
  - recruitment == "Crowdsourcing"              -> "Yes" or
      "No" only
  - recruitment == "Authors" / "Other" / "Mixed" / "Not
      reported" -> MUST be "NO"

### 16. annotator_training -- "Yes" or "No"
  - "Yes" if training, practice rounds, trial tasks,
      calibration sessions, or hands-on prior experience with
      the annotation tool are mentioned
  - "No" if no training is described, or if there is an
      explicit mention of no training provided
  - DEFAULT: "No"

### 17. language_proficiency -- PICK ONE exactly:
  * "Native"          (native speakers explicitly stated; OR
      annotators' ethnicity is reported and it aligns with
      the language of the annotation task)
  * "Non-Native"      (authors report some level of
      proficiency, e.g. fluent, B2, C1, but not native)
  * "Mixed"           (both native and non-native speakers
      were employed)
  * "Medium"          (annotators have moderate language
      proficiency, e.g. advanced learners)
  * "Not reported"    (language proficiency not directly
      reported)
  - Do NOT infer from dataset language or annotator country
      of residence alone.
  - DEFAULT: "Not reported"

### 18. expertise_level -- PICK ONE exactly:
  * "High"         (annotators have specialised domain
      knowledge relevant to the task, e.g. medical
      professionals, legal experts, trained linguists, NLP
      researchers, annotating linguistic phenomena)
  * "Medium"       (annotators have some relevant background
      but are not deeply specialised, e.g. students in a
      related field, people with self-reported relevant
      experience)
  * "Mixed"        (reported levels of expertise vary, e.g.
      experts and students, or students and crowd workers
      used together)
  * "Not reported" (the task COULD require specialised
      knowledge, but the paper does not specify annotators'
      expertise level)
  * "General Language/Knowledge task" (the task is
      straightforward and DOES NOT REQUIRE any specialised
      knowledge -- e.g. rating text
      fluency/relevance/acceptability using only common
      language sense, evaluating whether a headline is
      attractive, binary preference judgments on everyday
      content)
  - DEFAULT: "Not reported"

### 19. reported_compensation -- PICK ONE exactly:
  * "Paid"         (any compensation is mentioned: hourly
      rate, per-task payment, total amount, gift cards,
      course credit used as incentive, being described as a
      paid/professional annotator or employee)
  * "Free"         (annotation was done without payment:
      authors annotated the data themselves, annotators were
      volunteers or labelled as unpaid, students completed it
      as a course assignment with no monetary reward)

  * "Not reported" (the paper says nothing about how or
      whether annotators were compensated)
  - DEFAULT: "Not reported"

### 20. payment_rate -- PICK ONE exactly:
  * "Specific numeric rate" (a concrete payment is given:
      hourly rate like "$12/hr", per-task like "$0.05/HIT",
      total amount like "paid $500 total", or any explicit
```

```
      monetary figure)
  * "General mention"       (compensation is confirmed but no
      exact amount: "annotators were paid", "workers received
      payment", "full-time employees", "compensated for their
      time")

  * "NA"                    (reported_compensation is "Free"
      or "Not reported" -- payment information not available)
  - DEFAULT: "NA"
  [!] Check tables and appendices for payment figures --
      amounts are often reported there, not in the main text.

### 21-26. Socio-Demographics -- ALL are "Yes" or "No" ONLY
  - **age_reported**: "Yes" ONLY if actual age data (mean
      age, age range, age distribution) is reported WITH
      values. "No" otherwise.
  - **gender_reported**: "Yes" ONLY if actual gender data
      (counts, percentages, distribution) is reported WITH
      values. "No" otherwise.
  - **nation_origin_reported**: "Yes" ONLY if annotators'
      nationality/country of origin is reported WITH values.
      "No" otherwise.
  - **nation_residence_reported**: "Yes" ONLY if annotators'
      country of residence is reported WITH values. "No"
      otherwise. Indirect references to the residence of
      participants (e.g. poets from a local club, faculty
      employed at an American university department of X,
      English-speaking country) should be treated as Nation
      of residence = Yes.
  - **education_reported**: "Yes" ONLY if annotators'
      education level or field of study is reported WITH
      values. "No" otherwise.
    [!] INTERDEPENDENCY: If expertise_level == "High", set
      education_reported = "Yes" (explicit expert labelling
      implies education was assessed).
  - **political_orientation_reported**: "Yes" ONLY if
      political orientation is reported WITH values. "No"
      otherwise.
  *** CRITICAL DEFAULT: "No" for ALL demographic fields. ***
  *** "No" does NOT mean the paper says "we did not collect
      demographics". ***
  *** "No" simply means the paper does NOT report that
      specific demographic data. ***
  *** NEVER use "Not reported" for these fields -- only "Yes"
      or "No". ***

### 27. quality_control_filtering -- "Yes" or "No" ONLY
  - "Yes" if: quality control measures are described (related
      to discarding annotations or deboarding annotators:
      attention checks, gold or repeat questions, filtering
      bad annotators, removing low-quality annotations)
  - "No" in ALL other cases -- including when experts were
      employed and their judgments trusted without filtering,
      or when no QC details are given
  - *** NEVER output "NA" for this field -- "NA" is treated
      as "No", use "No" directly ***
  - DEFAULT: "No"

### 28. disagreement_resolution_adjudication -- PICK ONE
      exactly:
  * "Majority vote"      (final label determined by majority
      voting among annotators)
  * "Expert adjudication" (an expert resolves disagreements)
  * "Third annotator"     (a third annotator breaks ties)
  * "Consensus discussion" (annotators discuss to reach
      consensus)
  * "Soft labels / Distributional labels" (all annotations
      kept as a distribution, no single label chosen)
  * "Weighted voting"     (votes weighted by annotator
      reliability/expertise)
  * "Other / mixed"       (other resolution method or
      combination of methods)
  * "None"                (no disagreement resolution
      described: all opinions used as-is or averaged)
  - DEFAULT: "None"

## Extraction Rules -- READ CAREFULLY

### Experiment Identification

[!] **ANNOTATION EXPERIMENTS ONLY** -- Only extract
      experiments where humans actively create, label, rate,
      rank, or judge data. Do NOT extract model training
      runs, automatic evaluation steps, or system-based
      experiments that do not involve human annotators as a
      core part of the process.

**SPLIT into separate experiments when ANY of the following
      differ between annotation tasks:**
  1. **Different annotation objective** -- distinct label
      spaces or annotation goals (e.g. hate speech detection
      vs. stance detection)
  2. **Different annotator pools** -- different groups used
      for different tasks (e.g. experts for one subtask,
      crowd for another)
  3. **Different recruitment / compensation** -- one task
      paid, another unpaid; or recruited via different
      channels
  4. **Different annotation units** -- sentence-level vs
      document-level; word-level vs sentence-level
  5. **Separate dedicated annotation rounds** -- the paper
      describes multiple distinct rounds, each with its own
      guidelines, annotators, or IAA

Real-paper SPLIT examples:
    * binary attack classification / language classification
      / attack target classification -> 3 experiments
      (different label spaces)
    * summarization generation / summarization rating -> 2
      experiments (generation task + evaluation task)
    * Expert Gold-Standard NER / Crowd-Based NER -> 2
      experiments (different annotator pools)
    * poetry generation by crowd / poetry generation by
      experts -> 2 experiments (different recruitment)
    * Is label Acceptable/Unacceptable / If Acceptable, is it
      Precise/Imprecise -> 2 experiments (different
      annotation objectives)

**MERGE into one experiment when:**
  - The same annotation was applied to multiple datasets with
      the same label scheme and annotators
  - Multiple IAA metrics are reported for the same annotation
      task
  - Minor sub-tasks that share annotators, and annotated
      items, type of human judgment, and process differ only
      in topic or domain

[!] DEFAULT BIAS -- **ALWAYS err toward SPLITTING, not
      merging.** Under-splitting is the most common error. A
      paper that runs hate speech labeling and then
      target-type labeling on the same dataset has TWO
      experiments, not one. Different label spaces =
      different experiments.

[!] **SELF-AUDIT before writing JSON**: After identifying
      your first experiment, explicitly ask:
  1. Does the paper describe any OTHER annotation task
      (different labels, different annotators, different
      purpose in a different paper section)?
  2. Are there separate sections titled "Experiment 2",
      "Human Evaluation", "Quality Annotation", "Pilot Study"?
  3. Did any OTHER group of humans label, rate, or judge data
      in this paper?
  4. Is there a text production task AND a separate
      evaluation/rating task performed by humans?
  If the answer to ANY of these is yes -> add another
      experiment.

### CRITICAL: Avoid These Common Mistakes
1. **Do NOT use "Not reported" for demographic fields** --
      use "No". These fields ONLY accept "Yes" or "No".
2. **disagreement_resolution_adjudication must be one of the
      specific methods** -- "Majority vote", "Expert
      adjudication", "Third annotator", "Consensus
      discussion", "Soft labels / Distributional labels",
      "Weighted voting", "Other / mixed", or "None".
3. **Do NOT infer language proficiency** from the dataset
      language or annotator country alone (exception:
      ethnicity matching task language -> "Native").
4. **Do NOT infer compensation amounts** -- but DO infer
      "Free" when authors annotate their own data, or mention
      voluntary effort.
5. **Do NOT assume guidelines are released** just because the
      paper describes the task in prose -- there must be a
      dedicated document, appendix, URL, or cited source.
6. **Do NOT calculate workload fields** from other numbers --
      only extract explicitly stated values.
7. **For recruitment**, use the broad categories:
      "Crowdsourcing", "Authors", "Other", "Mixed", "Not
      reported" -- do NOT use platform-specific names.
8. **For compensation, do NOT use "NA"** -- use "Not
      reported" if compensation is unmentioned.
9. **Do NOT confuse expertise_level "General
      Language/Knowledge task" and "Not reported"**: "General
```

```
      Language/Knowledge task" = task needs no specialization
      at all; "Not reported" = task could need specialization
      but paper doesn't specify annotators' expertise.
10. **For IAA values, check tables** -- numeric values are
      very often in tables, not the prose. Do not output "NA"
      for a real metric until you have checked all tables.
11. **For total_annotated_items, use the annotated subset
      size**, not the total corpus/dataset size -- these can
      differ substantially.
12. **subtask_name must name the ANNOTATION task**, not the
      downstream NLP modelling task -- "sentiment rating" not
      "sentiment classifier"; "NER tagging" not "NER model".
13. **For every numerical value, record evidence** in the
      `numerical_evidence` object -- do NOT leave it empty
      when you extracted a non-"Not reported" value.

### Field Interdependencies -- APPLY THESE BEFORE FINALISING
      OUTPUT
These conditional rules take PRECEDENCE over all other
      guidance. Check each one after filling every field.

**A. iaa_value is fully determined by iaa_metric_name (apply
      per list position):**
  - iaa_metric_name[i] == "Other"        -> iaa_value[i] =
      "NA"
  - iaa_metric_name[i] == "Not reported" -> iaa_value[i] =
      "Not reported"
  - iaa_metric_name[i] is any real metric -> iaa_value[i] =
      the numeric value reported (or "NA" if value is absent)

**B. iaa_metric_name / iaa_value <- annotators_per_item or
      task type:**
  - annotators_per_item == "1" (single annotator)
                    -> iaa_metric_name = ["Not reported"],
      iaa_value = ["Not reported"]
  - task is open-ended text generation (humans produce free
      text, no fixed label set) -> iaa_metric_name = ["Not
      reported"], iaa_value = ["Not reported"]

**C. crowd_qualifications_screening <- recruitment:**
  - recruitment == "Crowdsourcing"   ->
      crowd_qualifications_screening = "Yes" OR "No" (based
      on paper content)
  - recruitment == "Authors"         ->
      crowd_qualifications_screening = "NO"
  - recruitment == "Other"           ->
      crowd_qualifications_screening = "NO"
  - recruitment == "Mixed"           ->
      crowd_qualifications_screening = "NO" (unless paper
      explicitly mentions crowd screening for the crowd
      sub-group)
  - recruitment == "Not reported"    ->
      crowd_qualifications_screening = "NO"

**D. education_reported <- expertise_level:**
  - expertise_level == "High" (experts)  ->
      education_reported = "Yes"
    (Rationale: explicitly labelling annotators as domain
      experts implies their educational/professional
      qualification was assessed and reported.)
  - All other expertise_level values     -> apply standard
      "Yes"/"No" rule

### Default values summary (when information is absent from
      the paper):
  - guidelines_released -> "No"
  - iaa_metric_name -> ["Not reported"]
  - iaa_value -> ["Not reported"]
  - workload fields -> "Not reported"
  - recruitment -> "Not reported"
  - crowd_qualifications_screening -> "NA" (non-crowd) or
      "No" (crowd)
  - annotator_training -> "No"
  - language_proficiency -> "Not reported"
  - expertise_level -> "Not reported"
  - reported_compensation -> "Not reported"
  - payment_rate -> "NA"
  - human_evaluated_sample -> "Not reported"
  - ALL demographics -> "No"
  - quality_control_filtering -> "No" (NEVER "NA")
  - disagreement_resolution_adjudication -> "None"

### Numerical Evidence
For every numerical value you extract, you MUST populate the
      `numerical_evidence` object with the verbatim text
      (sentence fragment, table cell, or figure caption) from
      the paper that supports that number.
  - **Fields that require evidence**: `annotators_per_item`,
      `total_annotators`, `total_annotated_items`,
      `items_per_annotator`, each element of `iaa_value`
      (keyed as `iaa_value_0`, `iaa_value_1`, etc.), and
      `payment_rate` when it is "Specific numeric rate".
  - If a value was inferred rather than directly stated,
      write: `"field_name": "inferred from: [brief reason]"`
  - Omit a key from `numerical_evidence` only when the
      corresponding field value is "Not reported" or "NA".


## Approach -- Two-step extraction

**STEP 1 -- Locate annotation sections BEFORE filling any
      field.**
Annotation metadata is typically found in sections titled:
      "Annotation", "Data Collection",
"Annotators", "Human Evaluation", "Experimental Setup",
      "Crowdsourcing", "Dataset Construction",
"Annotation Procedure", "Annotation Scheme", or appendix
      sections.
[!] Only extract HUMAN ANNOTATION experiments (humans
      labeling/rating/judging data). Ignore model training
      runs, automatic evaluation steps, or any experiment
      that does not involve human annotators.
Scan those sections and mentally note:
  * How many DISTINCT annotation experiments? (different
      tasks or different annotation units = separate
      experiments; otherwise merge)
  * Who are the annotators -- crowd workers, authors, domain
      experts, students?
  * How many annotators per item? Total annotators count?
  * What IAA metric(s) and numeric value(s) are reported?
  * How were annotators recruited and compensated?
  * Any demographics WITH VALUES, training, guidelines, or
      quality control described?
  * For each numerical value found, note the EXACT sentence
      or table entry it came from (for `numerical_evidence`).
**Multi-experiment scan**: Papers with 2-3 distinct
      annotation experiments are COMMON. After finding your
      first experiment, continue scanning:
  - Other sections / appendices for additional annotation
      tasks
  - Whether the paper reports a "pilot" or "validation"
      annotation with different setup
  - Whether different tasks/label-spaces were used (e.g.,
      first hate-speech, then target classification)
  - Whether two different annotator groups (crowd + expert)
      independently annotated the same or related data
  Do NOT stop at the first annotation section you find.

**STEP 2 -- Fill the experiments array** using only what you
      found in the paper.
When information is absent, apply the defaults in
      TAXONOMY_SCHEMA (typically "Not reported", "No", or
      "NA").

## WORKED EXAMPLE

Hypothetical paper: MTurk hate speech annotation with Fleiss'
      kappa, residence filter, majority-vote adjudication.

{{
  "paper_id": "example-paper",
  "skip": false,
  "experiments": [
    {{
      "paper_experiment_id": "example-paper-1",
      "papers_topic": ["Subjective language & social
      meaning"],
      "nlp_task_type": "Closed-label classification",
      "subtask_name": "hate speech detection",
      "intended_use": ["Resource creation"],
      "guidelines_released": "No",
      "iaa_metric_name": ["Fleiss' kappa"],
      "iaa_value": ["0.62"],
      "annotators_per_item": "3",
      "total_annotators": "50",
      "total_annotated_items": "10000",
      "items_per_annotator": "Not reported",
      "human_evaluated_sample": "Not reported",
      "recruitment": "Crowdsourcing",
      "crowd_qualifications_screening": "Yes",
      "annotator_training": "No",
      "language_proficiency": "Native",
      "expertise_level": "General Language/Knowledge task",
      "reported_compensation": "Paid",
```

```
      "payment_rate": "Specific numeric rate",
      "age_reported": "No",
      "gender_reported": "No",
      "nation_origin_reported": "No",
      "nation_residence_reported": "Yes",
      "education_reported": "No",
      "political_orientation_reported": "No",
      "quality_control_filtering": "Yes",
      "disagreement_resolution_adjudication": "Majority vote",
      "numerical_evidence": {{
        "annotators_per_item": "3/item",
        "total_annotators": "50 annotators from MTurk",
        "total_annotated_items": "collected 10,000 posts",
        "iaa_value_0": "Fleiss' kappa = 0.62"
      }}
    }}
  ]
}}

## Output Format
Output ONLY valid JSON matching this schema:
{json_schema}

## Paper Content
{paper_content}

## REMINDER before you write your JSON:
- **analysis**: write this FIRST -- 1-2 sentences: N
      experiments, annotator type, IAA metric+value,
      compensation
- Demographics (age/gender/nation_origin/
/education/political_orientation): default is "No", ONLY
      "Yes" if actual values WITH NUMBERS are reported
- disagreement_resolution_adjudication: use specific method
      name ("Majority vote", "Expert adjudication", "Third
      annotator", "Consensus discussion", "Soft labels /
      Distributional labels", "Weighted voting", "Other /
      mixed", "None")
- **IAA values**: check ALL tables -- values are very often
      in a table, not the prose
- recruitment: "Crowdsourcing" only if platform named or
      "crowd workers" used; "Authors" if authors themselves
      annotated; "Other" for students/experts/assistants;
      else "Not reported"
- **compensation**: authors annotating =
      reported_compensation "Free"; any pay mentioned =
      "Paid"; nothing mentioned = "Not reported"; "NA" if not
      applicable
- **payment_rate**: "Specific numeric rate" / "General
      mention" / "No" (if Free) / "NA" (if Not reported or
      NA) -- check tables for exact amounts
- **expertise_level**: "General Language/Knowledge task" if
      task needs no specialization (rating
      fluency/attractiveness/preference); "Not reported" if
      specialization COULD be relevant but paper doesn't
      specify
- **guidelines_released**: "Yes" if appendix with detailed
      instructions, URL, cited guidelines paper, or
      supplementary material -- NOT just because the paper
      describes the task
- **total_annotated_items**: use the annotated subset count,
      NOT the full corpus/dataset size
- **subtask_name**: use SPECIFIC wording mirroring the paper
      (e.g. "binary online attack classification", NOT
      "attack annotation"; "Pairwise Comparison of
      Model-Generated Summaries", NOT "evaluation")
- **papers_topic**: LIST of one or more topic areas
- **intended_use**: LIST of one or more intended uses
- **human_evaluated_sample**: free-text describing sample
      size/scope that was selected for human inspection or
      cross-annotated, or "Not reported"
- **Multiple experiments**: papers with 2-3 distinct
      annotation experiments are common -- re-read the FULL
      paper before settling on 1 experiment. Different label
      spaces, different annotator groups, generation +
      rating, pilot + main -> each is a SEPARATE experiment
      row
```

F Additional Analysis Details

RQ1: Additional analysis Papers on subjective and socio-linguistic phenomena do not exhibit systematically stronger reporting practices than the rest of the sample (see Figure 6). Regarding recruitment and qualification practices, socially grounded papers report annotator recruitment information slightly more often, suggesting more detailed documentation of annotator setup. These papers are also more likely to rely on crowdsourced and mixed annotator pools, and less likely to employ authors as annotators (see Figure 12).

As shown in Figure 7, with respect to language proficiency, papers on subjective language and social meaning topics report annotator nativeness more frequently than the rest of the dataset. A chi-square test indicates a statistically significant association between subset membership and native language reporting ($\chi^2 = 20.50, p < 0.001$), although the effect size is small (Cramer's $V = 0.088$), suggesting a weak association. At the same time, native speaker status is explicitly reported in fewer than 35% of socially grounded papers overall.

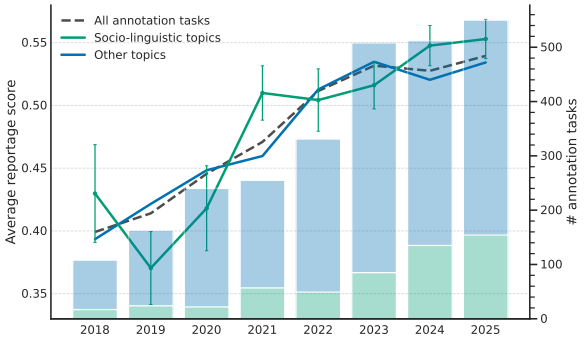


Figure 6: Trends in reporting quality for socio-linguistic vs. other topics. Lines show reporting improvement over time; stacked bars indicate the number of papers per category. Error bars represent standard errors of the mean (SEM) for the focus group only. Stacked bars in the background reflect the frequency of the socially grounded papers in ANNOTATED$_{\text{EXTRACT}}$.

It was observed that papers studying social phenomena are more likely to rely on crowdsourced and mixed annotator pools. Building on this higher use of crowdsourcing, crowd-annotated papers are also shown to exhibit a higher rate of quality control measures in the social subset compared to the rest of the dataset (see Figure 8).

This includes more frequent screening or filtering of crowdworkers in these studies. As shown in Figure 9, annotator training is also slightly more

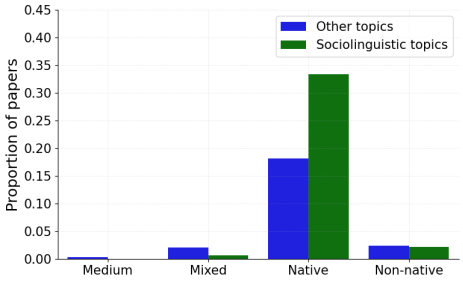


Figure 7: Language proficiency reporting rates, with higher reporting of nativeness for the socially grounded subset (green) compared to the rest of the dataset (blue).

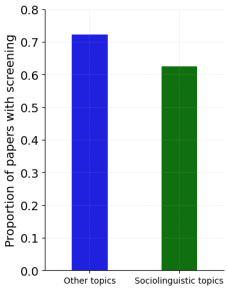


Figure 8: Distribution of quality control measures in crowd-annotated papers, defined as post-annotation filtering or validation of annotations, showing a higher rate in the socially grounded subset compared to the rest of the dataset (72% vs. 62.5% respectively).

frequently reported in the social subset. This difference is small but consistent, particularly when annotators belong to explicitly defined non-crowd groups (e.g., students, experts, hired assistants, or colleagues), or when multiple annotator groups are involved (see Figure 10).

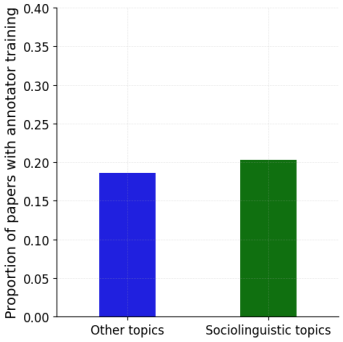


Figure 9: Annotator training reporting rates in socially grounded papers versus the full dataset, showing a slightly higher frequency in the socially grounded subset (20% vs. 18%), with an overall difference of approximately 2%.

Considering the category related to annotators' expertise, no evidence is found to support the hypothesis that papers addressing socio-linguistic topics rely more on expert annotators. In fact, these

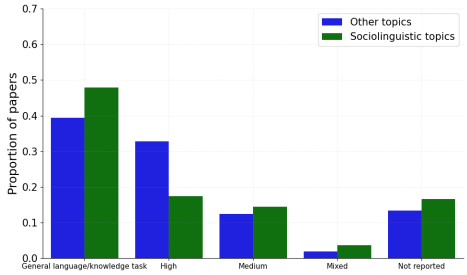


Figure 10: Distribution of annotator expertise in socially grounded papers versus the rest of the dataset, showing no higher use of expert annotators in the social subset, but a higher share of general language or knowledge-based tasks and more cases where annotator expertise is not reported.

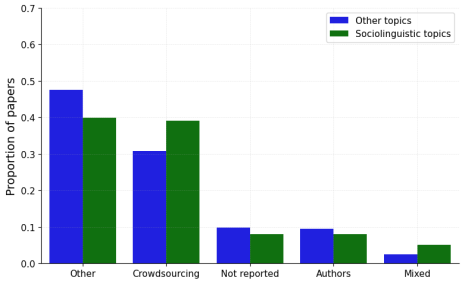


Figure 12: Distribution of recruitment categories in the socially grounded subset (green) versus the rest of the dataset (blue), showing higher use of crowdsourced and mixed annotator pools and slightly more frequent reporting of recruitment information in socially grounded papers.

studies include a higher proportion of tasks classified as general language- or knowledge-based, and more frequently do not report annotator expertise, as illustrated in Figure 10. This contrasts with the general pattern observed for other annotator-related dimensions, where papers on socio-linguistic topics tend to provide more detailed reporting. A further difference emerges in how annotation disagreements are handled in ordinal or scalar rating tasks. As shown in Figure 11, within the socially grounded subset, adjudication is more commonly performed through majority voting, followed by approaches that preserve disagreement as label distributions. In contrast, methods such as expert adjudication or consensus-based discussion are rarely employed. This suggests that, in addition to more detailed reporting, socially grounded studies tend to adopt aggregation strategies that retain or simplify annotator variation rather than resolving it through expert intervention.

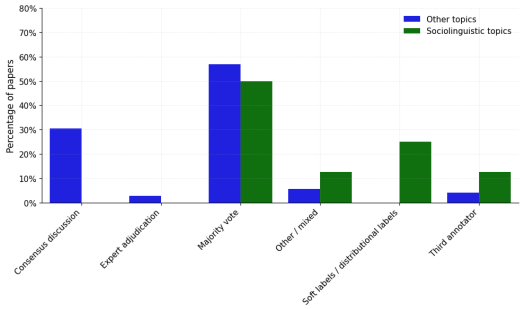


Figure 11: Distribution of adjudication strategies in ordinal and scalar rating tasks, showing that socially grounded papers more often use majority voting (50.0%) or label distributions (25.0%) that preserve disagreement, while rarely relying on expert adjudication or consensus-based discussion.

Overall, socially oriented NLP studies do not exhibit substantially different overall reporting quality, but they do show some distinctive tendencies in annotator profiles and documentation practices, including lower reliance on authors as annotators, greater use of crowdworkers, and more frequent reporting of native-speaker status. More broadly, these studies tend to document annotator characteristics, particularly annotator sourcing and language background, more systematically and in greater detail. However, reporting remains relatively low even within this subset, suggesting that socially grounded papers do not consistently document the broader methodological details needed to assess annotation quality, despite the importance of annotator-related variables for the phenomena they study.

RQ2: Venue-specific interrupted time series analysis As a heterogeneity check, we repeated the interrupted time series analysis in Section 6.2 separately for the three venues with the largest representation in our dataset: ACL, EMNLP, and NAACL. The interrupted time series analysis uses ordinary least squares (OLS), as implemented in the `statsmodels` library (formula: `score ~ time + post_2022 + time:post_2022`). The shaded bands in Figure 3 represent 95% confidence intervals for the mean REPORTAGE SCORE, computed using the conventional OLS covariance matrix and Student's t-distribution.

Figure 13 shows venue-specific fitted trends, observed yearly reportage scores, and counterfactual continuations of the pre-2022 trajectories.

The three venues exhibit distinct pre-2022 reporting trajectories. EMNLP shows comparatively high REPORTAGE SCORE values throughout the observation period, with no pronounced discontinuity around 2022 beyond the existing upward

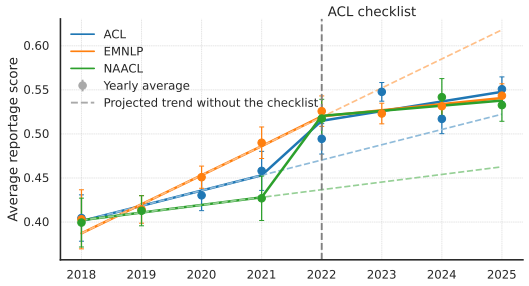


Figure 13: Interrupted time series analysis of reportage across major NLP venues (ACL, EMNLP, NAACL). Solid colored lines show model-estimated trends for each venue, while dots with whiskers indicate observed yearly means with standard errors. Dashed lines represent counterfactual trajectories under continuation of the pre-2022 trend.

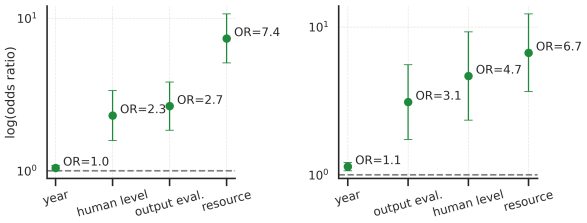
trend. In contrast, ACL and especially NAACL begin from substantially lower baseline levels and display stronger upward trends prior to 2022, gradually converging toward the EMNLP level.

Following 2022, the previously observed upward trajectories flatten across all three venues. Rather than accelerating post-intervention improvements, reportage scores stabilize at a more similar level across venues, reducing between-venue variation. This pattern suggests that the ACL Responsible NLP Checklist may have had a standardizing effect on reporting practices, even though we do not observe evidence for a strong immediate increase in reportage quality after its introduction.

Because the checklist was introduced during a broader transition period involving ARR adoption and venue-specific rollout schedules, the intervention year should be interpreted as an approximate temporal reference point rather than a perfectly synchronized policy change across all venues.

RQ3: Additional analysis Figure 14 visualizes the logistic regression estimates discussed in the main text. We fitted binary logistic regression models using the `statsmodels` library to predict the reporting of two sets of quality-related attributes. For panel (a), the binary outcome indicates whether a task reports at least one of the following: adjudication, post-annotation filtering, annotator training, or an IAA metric. Panel (b) analogously considers recruitment source and compensation. The predictors indicate whether the task is intended for resource creation, model-output evaluation, or benchmarking human performance, with publication year included as a control (formula: `has_qc ~ is_resource + is_model + is_human + year_num`). Parameters were estimated by maximum likelihood using the Newton–Raphson optimizer, and inference was based on two-sided Wald tests.

In both subplots, annotation tasks aimed at resource creation (resource) show the strongest positive association with reporting procedural details, with odds ratios substantially above 1 (dashed line) across all measures considered. Tasks aimed at benchmarking human performance (human level) occupy an intermediate position, whereas model-output evaluation tasks (output eval.) consistently show the weakest reporting patterns.



(a) QC measures combined (b) recruitment+comp.

Figure 14: Logistic regression of quality-control (QC) usage by annotation purpose and year. Points show odds ratios (ORs) with 95% CIs on a log scale. The dashed horizontal line at OR = 1 indicates no association. (a) QC usage indicates the presence of at least one of the following: adjudication, post-filtering, annotator training, or an IAA metric; (b) recruitment source and compensation are analyzed separately.

The similarity between the two panels suggests that these differences are not limited to formal quality control procedures alone (a), but extend more broadly to annotator transparency and documentation practices (b) – who are the annotators? Notably, the comparatively weak reporting of recruitment source and compensation in model evaluation studies raises the possibility that authors acting as annotators may remain systematically underreported in this setting. Temporal effects are comparatively modest in both models, indicating that annotation purpose is a substantially stronger predictor of reporting behavior than publication year.

Sample Annotated Data The following figures present excerpts from Haber et al. (2023), included as an example of a manually annotated paper with comprehensive reporting practices. The excerpts illustrate reporting of annotator recruitment, demographic and linguistic characteristics, workload distribution, inter-annotator agreement (IAA), and disagreement resolution procedures.

3.2 Data Annotation

We recruited a team of 14 annotators through Upwork, a crowdworking platform. All annotators were screened using a set of example annotation questions, and then onboarded and trained for our annotation task. We also asked all annotators to complete a short survey about themselves. 11 of the annotators are Indonesian, two are Malaysian and one declined to give this information. All annotators primarily resided in the country of their nationality. Annotators could enter their ethnicity as free-text. Two identified as Chinese, two as Malay, two as Gorontalo, and two as Asian. One identified as Minang, one as Javanese, one as being from Flores, and one from Sumatra. One annotator identified as mixed and another declined to give this information. All annotators were intermediate or fluent in English, and 12 were native, or near-native, in Indonesian. Six were intermediate or better in Malay, and many could speak other languages such as Japanese, Javanese, Korean and Tagalog. Ten of the annotators identify as women and four as men. Eight of the annotators are aged 18-29 years old and six are 30-39 years old.

Each annotator worked independently, labelling comments assigned to them according to extensive annotation guidelines based on our taxonomy of online attacks (§2). The annotators labelled whether comments contained an online attack or not, and if they did, they selected the target(s) of the attack. They also labelled the language(s) in which each comment was written. The 15,000 comments were annotated in six batches, with 10-14 annotators working on each batch. Annotation was *prescriptive* (Röttger et al., 2022), in the sense that we tasked annotators with applying the guidelines rather than their subjective beliefs. Each comment was annotated by three annotators. We make all

Figure 15: Example excerpt from Haber et al. (2023) reporting annotator information, including the number of annotators, recruitment procedures, and nation of origin.

Figure 16: Example excerpt from Haber et al. (2023) reporting annotation methodology and annotator characteristics, including language proficiency, gender, age, annotation counts, and annotators per item.

For the primary binary attack label, there was 3/3 agreement on 49.4% of comments, and 2/3 agreement on the rest. Fleiss' Kappa is 0.314.

For the language label, there was 3/3 agreement on 70.2% of comments. 3.9% of comments had three-way disagreement on the language label, which we resolved through expert annotation.

Figure 17: Example excerpt from Haber et al. (2023) reporting inter-annotator agreement metrics and disagreement resolution procedures.

category	value	2018-2021 #	2022-2025 #	2018-2021 %	2022-2025 %	p<0.05
paper_topic	open-ended generation	349	805	28.4	25.1	*
	semantics, inference & similarity	232	453	18.9	14.1	*
	subjective language & social meaning	120	423	9.8	13.2	*
	discourse & pragmatics	116	242	9.4	7.6	*
	qa & information access	108	265	8.8	8.3	
	information extraction	103	302	8.4	9.4	
	compression & reformulation tasks	95	201	7.7	6.3	
	multimodal tasks	44	247	3.6	7.7	*
	multilingual & translation	39	174	3.2	5.4	*
	linguistics	24	93	2.0	2.9	NA
task_type	closed-label clf	253	632	33.0	33.2	
	ordinal/scalar rating	229	551	29.9	29.0	
	open-ended generation	103	200	13.4	10.5	*
	comparative/preference	91	262	11.9	13.8	
	span-level/structural	68	156	8.9	8.2	
	other	22	100	2.9	5.3	NA
intended_use	model output evaluation	380	1043	47.1	50.5	
	resource creation	329	773	40.8	37.5	
	human performance	86	219	10.7	10.6	
	none of the above	11	29	1.4	1.4	NA
guidelines	no	612	1145	79.9	60.2	*
	yes	154	756	20.1	39.8	*
iaa_metric	not reported	502	1103	55.7	48.3	*
	fleiss' κ	116	272	12.9	11.9	
	cohen's κ	79	268	8.8	11.7	*
	krippendorff's α	68	289	7.5	12.7	*
	other	42	140	4.7	6.1	

category	value	2018-2021 #	2022-2025 #	2018-2021 %	2022-2025 %	p<0.05
	total agreement	37	124	4.1	5.4	
	spearman	21	23	2.3	1.0	NA
	f1 agreement	15	24	1.7	1.1	NA
	pearson	13	19	1.4	0.8	NA
	kendall's τ	5	19	0.6	0.8	NA
	majority agreement	3	3	0.3	0.1	NA
iaa_value	reported	765	1894	99.9	99.6	
	not_reported	1	7	0.1	0.4	NA
annotators/item	reported	528	1201	68.9	63.2	*
	not_reported	238	700	31.1	36.8	*
total_annotators	not_reported	387	601	50.5	31.6	*
	reported	379	1300	49.5	68.4	*
total_items	reported	673	1621	87.9	85.3	
	not_reported	93	280	12.1	14.7	
items/annotator	not_reported	640	1421	83.6	74.8	*
	reported	126	480	16.4	25.2	*
recruitment	crowdsourcing	358	476	46.7	25.0	*
	other	247	1009	32.2	53.1	*
	not reported	100	157	13.1	8.3	*
	authors	51	200	6.7	10.5	*
	mixed	10	59	1.3	3.1	NA
crowd_screening	no	492	1244	64.2	65.4	
	yes	186	341	24.3	17.9	*
	na	88	316	11.5	16.6	*
training	no	667	1502	87.1	79.0	*
	yes	99	399	12.9	21.0	*
lang._proficiency	not reported	631	1403	82.4	73.8	*
	native	116	390	15.1	20.5	*
	non-native	11	54	1.4	2.8	NA
	mixed	5	48	0.7	2.5	NA
	medium	3	6	0.4	0.3	NA
expertise_level	general task	448	613	58.5	32.2	*
	high	147	708	19.2	37.2	*
	not reported	108	253	14.1	13.3	
	medium	56	280	7.3	14.7	*
	mixed	7	47	0.9	2.5	NA
compensation	not reported	479	697	62.5	36.7	*
	paid	218	913	28.5	48.0	*
	free	69	291	9.0	15.3	*
payment_rate	na	548	988	71.5	52.0	*
	specific numeric rate	139	605	18.1	31.8	*

category	value	2018-2021 #	2022-2025 #	2018-2021 %	2022-2025 %	p<0.05
	general mention	79	308	10.3	16.2	*
age	no	760	1763	99.2	92.7	*
	yes	6	138	0.8	7.3	NA
gender	no	758	1744	99.0	91.7	*
	yes	8	157	1.0	8.3	NA
nation_origin	no	762	1845	99.5	97.1	*
	yes	4	56	0.5	2.9	NA
nation_residence	no	675	1621	88.1	85.3	
	yes	91	280	11.9	14.7	
education	no	596	1007	77.8	53.0	*
	yes	170	894	22.2	47.0	*
political_orient.	no	765	1887	99.9	99.3	
	yes	1	14	0.1	0.7	NA
post_filtering	no	551	1447	71.9	76.1	*
	yes	215	454	28.1	23.9	*
adjudication	none	593	1437	77.4	75.6	
	majority vote	108	228	14.1	12.0	
	consensus discussion	23	105	3.0	5.5	NA
	expert adjudication	14	42	1.8	2.2	NA
	other/mixed	14	49	1.8	2.6	NA
	third annotator	8	27	1.0	1.4	NA
	soft labels	3	11	0.4	0.6	NA
	weighted voting	3	2	0.4	0.1	NA

Table 7: Impact of the ACL Responsible NLP Checklist. Distributions are shown for 2018–2021 vs. 2022–2025. Multilabel fields are expanded into individual categories; free-text fields are binarized (reported vs. not reported). Statistical significance (*) reflects differences in proportions between periods, assessed using a χ^2 test ($p < 0.05$); results are omitted (NA) when counts fall below 30 in either period. Total observations (tasks) : 2,669.